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City2Graph: A Python library for Heterogeneous Graph Neural Networks and spatial analysis in urban systems[☆]

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ABSTRACT

City2Graph is an open-source Python library that streamlines workflows for heterogeneous Graph Neural Networks (GNNs) in urban systems. Cities are complex systems of diverse spatial relations long modelled as graphs in network science, and recent advances in GNNs have further enabled the identification of non-linear patterns of urban complexity. Unlike homogeneous graphs with a single node and edge type, heterogeneous graphs with multiple types are receiving growing attention to accommodate richer information in GNNs. However, their diffusion remains constrained by fragmented graph construction processes across different data domains, and by the lack of a unified framework for converting constructed graphs into GNN-ready tensors. City2Graph standardises graph construction across domains of morphology, transportation, mobility, and proximity. The library supports conversions between spatial geometries, network topologies, and tensors for spatial analysis with visualisation, network analysis, and GNN training, respectively. City2Graph also supports metapath construction, capturing higher-order connections across node and edge types (e.g., areas linked via multimodal transit). The library's efficacy was demonstrated through a case study on clustering urban functions in Liverpool, UK, using Graph Autoencoder models with three relation types: spatial contiguity, walk-based accessibility, and multimodal accessibility between census units. Compared with the homogeneous model, heterogeneous models identified spatially coherent clusters that aligned more closely with defined accessibility patterns. City2Graph enables reproducible workflows for heterogeneous GNNs and model interpretation, fostering comprehensive understanding of urban systems across disciplines. The library is released under the BSD 3-Clause License on GitHub.

1. Introduction

For more than a decade, cities have often been considered complex systems, where diverse entities interact through their spatial relations (Batty, 2013). When relations are discretely observed, urban systems are modelled as graphs, with entities (e.g., zones) as nodes and their connections (e.g., human flows between zones) as edges (Barthélemy, 2011). Such graph-based representations have become fundamental across urbanisation (Batty, 2008), migration (Fagiolo & Mastrorillo, 2013), transportation and mobility (Chang et al., 2021; Gallotti & Barthelemy, 2015), and urban morphology (Boeing, 2021; Hillier, 1996). Graph representation has been used to reveal the process of micro-level interactions that shape emergent patterns in the meso- and macro-scale, from metrics of network science such as node centralities and degree distribution. In particular, small-world (Watts & Strogatz, 1998) and scale-free (Barabási & Albert, 1999) theories have provided a framework for understanding the structural complexity of urban systems (Ducruet & Beauguitte, 2014; Masucci et al., 2014; Neal, 2018; Rozenblat & Melançon, 2013).

While graphs provide the conceptual foundation for modelling discrete urban relations, graph representation learning provides the methodological framework to exploit such structures in machine learning. Conventional models treat data as a table of independent rows or as an image on a regular pixel grid, so relations between observations are either ignored or fixed in advance. Graph representation learning instead learns from this irregular relational structure and expresses it as embeddings, which are vectors in Euclidean space assigned to nodes, edges, or whole graphs (Hamilton et al., 2017), as illustrated in Fig. 1. For example, given a graph of administrative areas as nodes and their spatial contiguity as edges, the node embedding of each area summarises both its own attributes (e.g., population or land use) and

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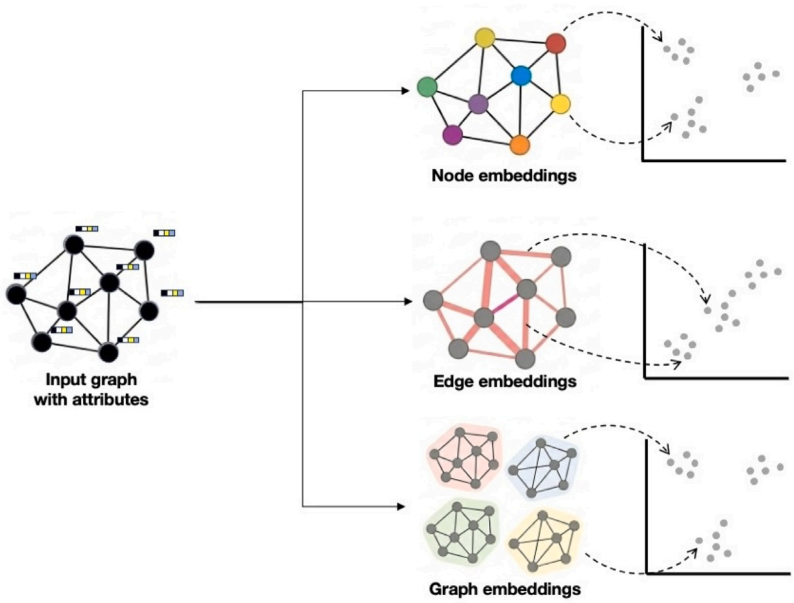


Fig. 1. Conceptual diagram of graph representation learning.

those of its neighbours, so that areas with similar surroundings lie close together in the vector space and become directly comparable for downstream tasks such as clustering, classification, and prediction.

Among graph representation learning approaches, Graph Neural Networks (GNNs) have gained growing attention. In the standard architecture of GNNs, each node iteratively updates its embedding after aggregating information from its neighbouring nodes along the edges (Khemani et al., 2024; Zhou et al., 2020). Stacking several such layers allows the model to consider broader areas in the graph. Through this mechanism, GNNs have outperformed conventional statistical methods in classifying land use (Gong et al., 2024; Xu et al., 2022), urban functional regions (Zhang et al., 2024), geodemographic properties (De Sabbata & Liu, 2023), and urban form homogeneity (Xue et al., 2022). By leveraging graphs that inherently encode spatial relative positioning, graph representation learning via GNNs offers a novel, spatially explicit modelling approach within the geospatial artificial intelligence (GeoAI) framework (Janowicz et al., 2020; Liu & Biljecki, 2022; Mai et al., 2022; VoPham et al., 2018).

Applying GNNs to urban systems benefits from a graph representation that can holistically capture the heterogeneous relations in cities. Unlike homogeneous graphs, which are limited to a single entity type, heterogeneous graphs model multiple node and edge types within a unified structure (Kivelä et al., 2014; Yang et al., 2022), capturing diverse semantic relations across physical and functional layers (De Domenico et al., 2015). In a heterogeneous graph, nodes could represent intersections, bus stations, amenities, and administrative areas, while edges encode their relations such as “connected by street segment”, “served by bus line”, or “within 15-minute walking distance”. Owing to this flexibility in design, heterogeneous graphs enable GNNs to learn from the multi-layered context of urban systems.

However, applying these methods to urban systems is constrained by technical challenges, notably the difficulty of constructing reproducible heterogeneous graphs from diverse spatial data with a fragmented software ecosystem. To address such challenges, we have developed City2Graph, a Python library designed for reproducible, interpretable heterogeneous graph representation learning in urban systems. As illustrated in Fig. 2, the library is built on two design principles. First, it provides pipelines to construct a standardised graph structure from urban data of multiple domains. Second, with the constructed graphs, it allows smooth conversions between different data objects while ensuring spatial alignment and semantic consistency, depending on the purposes of spatial analysis, network analysis, and GNNs.

Such conversions streamline workflows from feature engineering to model interpretation. On the input side, City2Graph enables users to enrich node, edge, and graph features with spatial indicators (e.g., density) and topological metrics (e.g., centrality), which are convertible to tensors for GNN training. On the output side, it facilitates inverse mapping of GNN outputs (e.g., embeddings and attention weights) back to original spatial entities, preserving geometries and coordinate systems. This restoration of spatial context enables exploratory spatial data analysis (ESDA) to interpret and visualise model results. City2Graph provides a robust foundation for researchers needing spatially explicit and explainable GeoAI modelling.

The remainder of the paper is as follows. Section 2 reviews related works. Section 3 presents City2Graph’s design and functionalities. Section 4 demonstrates the library through a case study on functional cluster identification in Liverpool, grouping census units by shared urban functions and relations of contiguity, walk-based accessibility, and multimodal accessibility between census Output Areas. Section 5 discusses implications and limitations, and Section 6 concludes.

2. Related works

2.1. Graph representations of urban systems

Graph representations provide a structural view of the relational complexity of urban systems, which arises from the non-linear interrelations among physical infrastructure, mobility flow, and socio-economic transactions (Batty, 2013; McPhearson et al., 2016). Analytical measures in network science, such as centrality, modularity, and degree distributions, have been widely applied across quantitative geography, ranging from identifying the local and global integration of urban spaces (van Nes & Yamu, 2021; Porta et al., 2012) to assessing topological robustness as urban resilience against failures (Akbarzadeh et al., 2019; Boeing & Ha, 2024; Kirkley et al., 2018; Sharifi, 2019) and migration pattern identification (Danchev & Porter, 2018; Davis et al., 2013). Beyond direct topological analysis, graph representations also serve as a theoretical foundation for spatial statistical methods, such as defining neighbourhood matrices for spatial autocorrelation and gravity models (LeSage & Polasek, 2008).

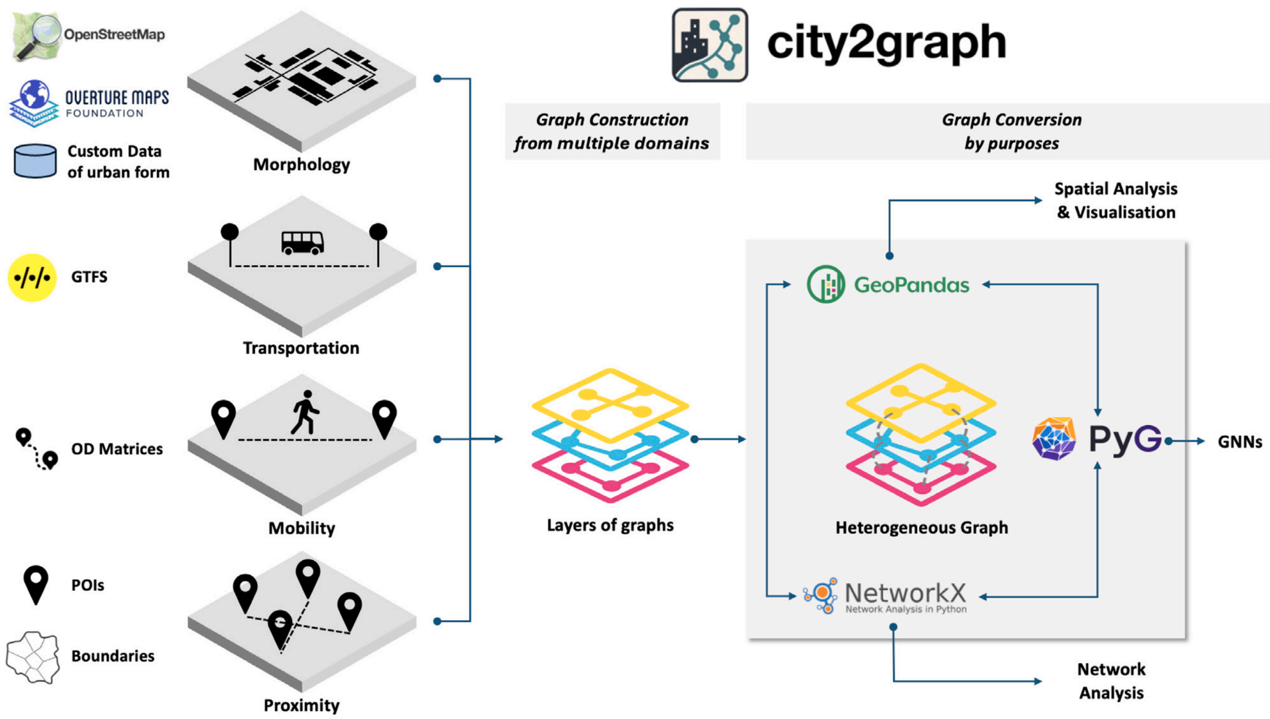


Fig. 2. Conceptual diagram of the City2Graph framework.

In network science, heterogeneous graphs have been employed through multilayer and multiplex network theory, which explicitly models both within-layer structure and inter-layer coupling (De Domenico et al., 2015; Halu et al., 2014; Kivelä et al., 2014; Kurant & Thiran, 2006). In quantitative geography, they have been applied to identify complex urban patterns, such as urban structure including polycentricity (Burger et al., 2014; Zhang et al., 2021), multimodal transport dynamics (Alessandretti et al., 2023; Jia et al., 2022; Li et al., 2024), transport system vulnerabilities (Aleta et al., 2017), migration patterns from multiple social networks (Belyi et al., 2017), and urban segregation (Neira et al., 2024).

Heterogeneous graphs are also applied to GNNs, integrating different node and edge types with varying feature dimensions (Hu et al., 2020; Wang et al., 2019; Zhang et al., 2019). Such comprehensiveness in modelling has been demonstrated across several urban domains. In functional region classification, Zhang et al. (2024) integrated metro passenger flow with points of interest (POIs) and building-level urban form, treating street delineation blocks and stations as distinct node types. Chen et al. (2024) took a different approach to traffic prediction, defining two relation types between sensors: one via street connectivity and the other via similarity of surrounding POIs. Land use inference has also benefited from heterogeneous designs. Gong et al. (2024) encoded three relations between zones based on spatial contiguity, inward human flows, and outward human flows. Liu et al. (2022) constructed heterogeneous GNNs from land coverage raster data using superpixels as nodes and combinations of coverage types as relations.

Across urban domains, integrating multiple data layers has improved the model performance. Liu et al. (2025b), for instance, combined street networks with waterway systems in a heterogeneous Graph Autoencoder, improving predictions of land surface temperature and public housing prices in Singapore by approximately 20% over street-only models. Kim and Yoon (2025) adopted a comparable strategy for socio-economic indicators in New York City, constructing a heterogeneous information network that integrated census tracts with taxi mobility flows and Foursquare POI check-ins. Zhai et al. (2025) combined multimodal mobility data from subway, bus, and bike-sharing networks to improve both accuracy and interpretability through feature attribution. At a different spatial scale, Liu et al. (2025a) proposed a hierarchically nested structure of census units as a heterogeneous graph, enhancing estimates of population and air pollution exposures in London from coarser units.

2.2. Open science and data sources for urban graph analysis and learning

Nowadays, urban researchers benefit from the ecosystem of open science outcomes, such as open data sources and open-source software (OSS). A diverse ecosystem of data sources now encodes potential graph structures, including urban morphology and POIs in OpenStreetMap (OSM) (OpenStreetMap contributors, 2026) and Overture Maps (Overture Maps Foundation, 2026b), transit schedules in General Transit Feed Specification (GTFS) format (Fayyaz et al., 2017), and mobility records (Zhao et al., 2016). The Python ecosystem has matured alongside such data pipelines. GeoPandas (Van den Bossche et al., 2025) provides robust spatial processing, NetworkX (Hagberg et al., 2008) supports general network analysis, and specialised libraries such as OSMnx offer standardised workflows for acquiring and analysing street networks (Boeing, 2017, 2025). For GNNs, PyTorch Geometric (Fey & Lenssen, 2019) enables the conversion of data to tensors and modelling with GPU support.

Table 1 compares open-source tools for spatial graph construction by the main use case and supported data domain. A variety of libraries covers tasks for their analytical purposes, including OSMnx (Boeing, 2017, 2025) and Urbanity (Yap et al., 2023) for street network analysis; Pandana (Foti et al., 2012), UrbanAccess (Blanchard & Waddell, 2017), and Madina (Sevtsuk & Alhassan, 2025) for accessibility modelling; and libpysal (Rey & Anselin, 2010; Rey et al., 2022) for spatial weights. Most of them, however, assume homogeneous graph structures or limited combinations of graph layers. Integration with GNN workflows is also constrained, where only STM-Graph (Ghaffari et al., 2025) currently supports direct data conversion.

As a summary, several research gaps can be identified. Constructing reproducible heterogeneous graphs from diverse spatial datasets remains difficult, as each data domain carries distinct schemas, spatial granularities, and temporal resolutions (Al-Yadumi et al., 2021).

Table 1
Comparison of open-source Python packages for spatial graph construction and their capabilities.

Tool	Primary Use Case	Graph Data Domain				Graph Data Sources	Graph Type	Conversion for GNNs	Latest Release on PyPI[a]	Reference
		Morph.	Trans.	Mob.	Prox.					
libpysal	Construction and analysis of spatial weights	–	–	–	✓ (Area/Point)	Custom polygons/points	Hom.	–	Jan. 9, 2026	Rey and Anselin (2010), Rey et al. (2022)
Pandana	Accessibility analysis with fast computation	✓ (General*)	✓ (General*)	✓ (General*)	✓ (General*)	Custom points/edges	Hom.	–	Jul. 26, 2023	Foti et al. (2012)
OSMnx	Street network acquisition and analysis for OSM data	✓ (Streets)	–	–	–	OSM	Hom.	–	Feb. 16, 2026	Boeing (2017, 2025)
UrbanAccess	Integrated pedestrian and transit accessibility analysis	✓ (Streets)	✓ (Transit)	–	–	OSM, GTFS	Het. (Street & transit distinguished by flags)	–	Nov. 9, 2020	Blanchard and Waddell (2017)
spaghetti	Spatial analysis with topological inferences	✓ (General*)	✓ (General*)	✓ (General*)	✓ (General*)	Custom linestrings	Hom.	–	Jun. 21, 2024	Gaboardi et al. (2021)
Cityseer	Pedestrian-scale urban form analysis	✓ (Streets)	✓ (Transit)	–	–	OSM, GTFS	Het. (Street & transit distinguished by flags)	–	Mar. 19, 2026	Simons (2023)
Urbanity	Multi-dimensional urban network analysis (feature engineering on street networks)	✓ (Streets)	–	–	–	OSM	Hom.	–	Jan. 31, 2026	Yap et al. (2023)
Madina	Accessibility analysis and flow simulation of pedestrian and bicycle trips	✓ (Streets)	–	✓ (OD)	–	OSM, OD	Het. (Streets & OD managed by layers)	–	–	Sevtsuk and Alhassan (2025)
STM-Graph	Spatio-temporal mapping and GNN modelling	–	–	–	✓ (Area)	Custom polygons	Hom.	✓	Jun. 1, 2025	Ghaffari et al. (2025)
neatnet	Simplification of street networks	✓ (Streets)	–	–	–	Custom linestrings	Hom.	–	Nov. 20, 2025	Fleischmann et al. (2026)

Note: Morph.: Morphology; Trans.: Transportation; Mob.: Mobility; Prox.: Proximity. Hom.: Homogeneous; Het.: Heterogeneous. ✓ indicates support. * means the use is domain-free for the supported geometries.
[a] As of Mar. 20, 2026.

The toolkit ecosystem compounds this challenge. Individual libraries are powerful in isolation, yet seamless conversion of data structures between spatial, network, and GNN frameworks is still limited for heterogeneous graphs, forcing researchers into ad hoc workflows that erode reproducibility. As the standard GNN architecture holds black-boxed nature with respect to the parameter behaviours (Liu et al., 2024), the absence of integrated pipelines hinders the interpretation of GNN outcomes such as embeddings and attention weights, restricting both explainability and uptake in policy-making. A unified framework is therefore needed, which converts heterogeneous data sources into standardised structures and bridges GNN outputs with established spatial statistics for validation.

3. City2Graph: design and functionalities

This section introduces City2Graph, an open-source Python library for constructing heterogeneous graphs for GNNs across domains of morphology, transportation, mobility, and proximity. City2Graph addresses fragmented workflows by standardising the pipeline from raw data preprocessing to graph construction and conversion between GeoDataFrames of nodes and edges, NetworkX objects, and PyTorch Geometric tensors. It also supports model interpretation by mapping GNN outputs such as embeddings back into geospatial objects for visualisation and spatial analysis. City2Graph can also be used for homogeneous GNNs, as well as spatial network analysis that does not involve GNN components.

While independent from any data sources, City2Graph offers dedicated pipelines for OSM and Overture Maps. Compared to OSM as a volunteer-based open data source, Overture Maps enhances the consistency of OSM by integrating other resources (Ochoa-Ortiz & Re, 2025; Overture Maps Foundation, 2026a). For OSM, the library ensures full compatibility with OSMnx-generated GeoPandas and NetworkX objects (Boeing, 2017, 2025), leveraging OSMnx's robust data acquisition while enabling integration with other domains. For Overture Maps, an open initiative led by the Overture Maps Foundation, City2Graph facilitates access to the datasets such as building footprints, street segments, and POIs. Further details on Overture Maps appear in Section 3.4. City2Graph is hosted on a GitHub repository at https://github.com/c2g-dev/city2graph under the BSD 3-Clause License. The library can be installed via PyPI and conda-forge. Official API documentation, installation instructions, and usage examples are available at https://city2graph.net.

Table 2 summarises the graph-construction components of City2Graph introduced in this section and the analytical workflows they

Table 2
Summary of the functions for graph construction in City2Graph.

Section	Major functions	Main behaviour	Spatial assumptions of the graph representation	Use case scenarios
Morphology (Section 3.1.1)	`segments_to_graph()`, `morphological_graph()`	Builds a homogeneous graph from any data source, with intersections as nodes and street segments as edges. Alternatively, builds a heterogeneous graph that distinguishes street segments as *movement* nodes from buildings and plot systems as *place* nodes, preserving three relations by their arrangement: *movement-to-movement*, *place-to-place*, and *place-to-movement*.	Treats streets (*movement*) and plots/buildings (*place*) as functionally distinct node types. Contiguity between plots approximates block-scale development patterns, while *place-movement* links through frontages approximate the connectivity of the whole urban system. The *movement* versus *place* role distinction is itself the object of analysis.	Street network analysis on OSM, Overture Maps, or custom data (`segments_to_graph()` with `gdf_to_nx()`). Assessment of 15-Minute City conformity via clustering of neighbourhood typologies that jointly consider functional diversity and walking access (`morphological_graph()` with `gdf_to_pyg()` and GNNs).
Transportation (Section 3.1.2)	`load_gtfs()`, `travel_summary_graph()`	Parses GTFS zip files and attaches spatial geometries to stops and routes, then builds a weighted stop-to-stop graph whose edges carry service frequency and average travel time for a specified calendar term and time range.	Defines connections by scheduled service rather than spatial proximity, representing reachability between stops and the level of service (frequency and travel time). The relations are time-dependent, conditioned on the specified calendar term and time range.	Identification of underserved time bands by comparing service frequency and travel time (`load_gtfs()`, `travel_summary_graph()`). Identification of modal transport inequality across an integrated walk, bus, and railway system (`travel_summary_graph()` with `segments_to_graph()`, `bridge_nodes()`, and `create_isochrone()`).
Mobility (Section 3.1.3)	`od_matrix_to_graph()`	Converts an origin–destination matrix, given as an edge list or adjacency matrix, into a graph whose nodes are zones (polygons or points) and whose edges are flows. Zone identifiers are preserved so that attributes can be attached in GeoPandas.	A flow-based representation that treats observed movement volumes as the strength of relations between zones. It presupposes neither spatial proximity nor physical connection; observed, and possibly asymmetric, interactions define the edges.	Identification of functional catchments through community detection on OD flows (`od_matrix_to_graph()` with `gdf_to_nx()` and network analysis). Estimation of unobserved OD and post-land-use flow changes through flow prediction using zonal attributes such as population, employment, and land use (`od_matrix_to_graph()` with `gdf_to_pyg()` and GNNs).
Proximity (Section 3.1.4)	`knn_graph()`, `fixed_radius_graph()`, `delaunay_graph()`, `gabriel_graph()`, `relative_neighborhood_graph()`, `euclidean_minimum_spanning_tree()`, `waxman_graph()`, `contiguity_graph()`, `bridge_nodes()`, `group_nodes()`	Generates edges from proximity relations between points, polygons, and across layers. For points, provides the parametric *k*NN and fixed-radius graphs, the non-parametric Delaunay, Gabriel, relative neighbourhood, and Euclidean minimum spanning tree graphs, and the distance-decay Waxman graph. For polygons, provides Queen/Rook contiguity; for connections across layers, bridge and group methods. Distance may be Manhattan (L1), Euclidean (L2), or network distance derived from linestrings.	Proxies unobserved relations through geometric closeness. *k*NN encodes relative proximity (rank), fixed-radius the physical meaning of absolute distance, and contiguity a shared boundary, so the choice of method is itself a hypothesis about the relevant spatial relationship.	Identification of amenity clusters through community detection on POI proximity graphs (`knn_graph()`, `fixed_radius_graph()` with `gdf_to_nx()` and network analysis). Geodemographic classification incorporating neighbourhood effects from the contiguity of administrative boundaries (`contiguity_graph()` with `gdf_to_pyg()` and GNNs). Construction of heterogeneous graphs that traverse separate layers (`bridge_nodes()`, `group_nodes()`).
Metapath (Section 3.1.5)	`add_metapaths()`, `add_metapaths_by_weight()`	Reduces a specified sequence of relations on a heterogeneous graph into direct semantic edges that aggregate a cumulative cost such as travel time. `add_metapaths_by_weight()` projects edges by Dijkstra's algorithm under a cumulative-cost threshold.	Abstracts multi-step pathways into direct higher-order relations. Relations can be defined not only by explicit specification (e.g., amenity → street → street → amenity), but also by a cumulative-cost threshold (e.g., amenity → (15-minute travel time) → amenity).	Construction of a functional proximity network that bypasses street and stop nodes by reducing multi-step walking and transit routes into direct facility-to-facility edges aggregating travel time, identifying hub facilities reached in common from many residences or jointly used amenity clusters (`add_metapaths()` with `travel_summary_graph()`, `bridge_nodes()`, and network analysis). Typological classification of urban function composition based on 15-minute walking reachability between census areas (`add_metapaths_by_weight()` with `morphological_graph()`, `bridge_nodes()`, `gdf_to_pyg()`, and GNNs; detailed in the case study, Section 4).

support. Across all domains, the choice of graph representation is not a neutral preprocessing step but encodes an assumption about which spatial relationship is treated as analytically relevant, recorded in the spatial-assumptions column of the table.

3.1. Graph construction

Graph construction in City2Graph is built on a standardised GeoPandas-based data structure (Van den Bossche et al., 2025) that is shared across all domains (morphology, transportation, mobility, and proximity). Every function returns graphs in consistent structures: homogeneous graphs are represented as a tuple of two GeoDataFrames `(nodes, edges)`, while heterogeneous graphs are represented as a tuple of two dictionaries `(nodes_dict, edges_dict)` whose values are GeoDataFrames. In heterogeneous graphs, `nodes_dict` has keys of node type (e.g., `'street'`), while `edges_dict` is keyed by edge type triplets `(source_type, relation, target_type)` (e.g., `('street', 'is_connected_to', 'street')`). This standardisation realises cross-domain operations such as the construction of metapaths (Section 3.1.5) and the conversion of heterogeneous graphs into tensors for GNNs (Section 3.2).

3.1.1. Morphology

The morphology module provides two graph construction methods. The `segments_to_graph()` function builds a primal graph, where intersections are nodes and street segments are edges, capturing their physical topology as a homogeneous graph. While effective for characterising movement channels, this representation disregards other fundamental urban forms in cities, such as the building footprints and plot systems or parcels (Conzen, 1960; Oliveira, 2016). Such non-street elements function as places that accommodate socio-economic activities, whose spatial arrangement through contiguity has the potential to describe the development patterns at the city block level (Bobkova et al., 2021; Krüger, 1979, 1980). Moreover, their integration with the

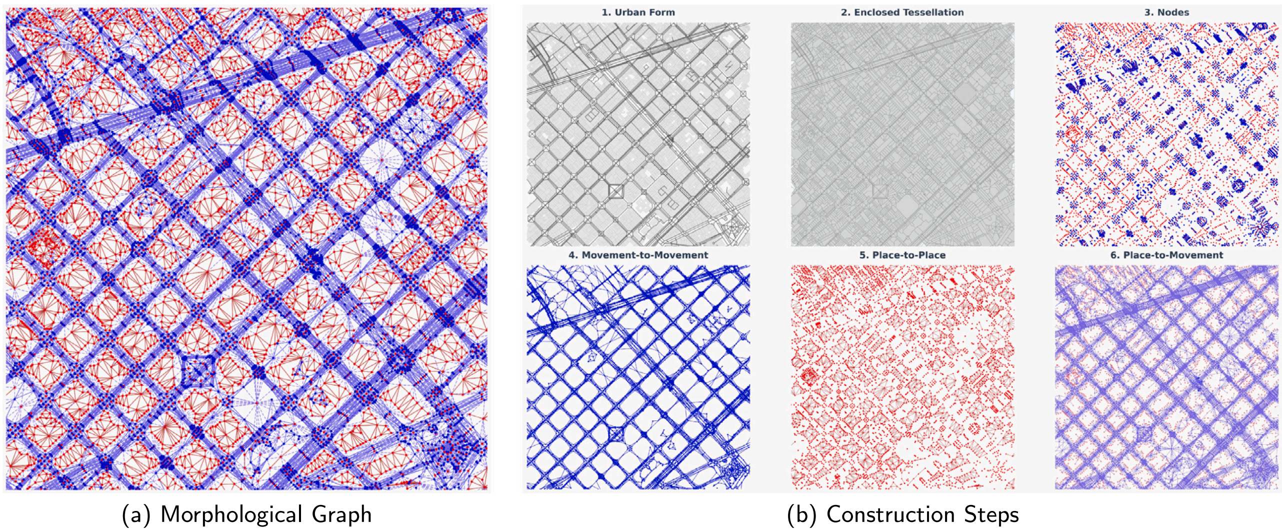


Fig. 3. Visualisation of (a) morphological graph and (b) its construction process: 1. Urban forms (building footprints and street segments), 2. Enclosed tessellation, 3. Nodes (Blue: *Movement*, Red: *Place*), 4. *Movement-to-movement* edges, 5. *Place-to-place* edges, 6. *Place-to-movement* edges. Data were extracted from Overture Maps (accessed on Jan. 17, 2026) for the extent of the Eixample district of Barcelona, Spain.

street network through frontages enables the connectivity of the entire urban system to be analysed (Domingo et al., 2019).

The `morphological_graph()` function addresses this limitation by synthesising a heterogeneous graph of street segments and plot systems with building footprints. Within this graph, urban form elements are distinguished by the role they play in the urban system: street segments serve as the channels through which movement is organised, whereas building footprints and their enclosing plot units serve as the units on which socio-economic activities and functions are located. Construction proceeds in two phases. In the first phase, raw geometries are transformed into an enclosed tessellation as an approximation of plot systems (Arribas-Bel & Fleischmann, 2022), providing tessellation cells as the *place* units and street segments as the *movement* channels of the resulting graph. In the second phase, topological connectivity is established through three relations. *Movement-to-movement* edges connect street segments via a dual graph approach (Masucci et al., 2014), preserving linear continuity and flow. *Place-to-place* edges link adjacent plot systems by spatial contiguity. *Place-to-movement* edges connect plots to their bordering street segments. As shown in Fig. 3, the resulting graph provides a comprehensive representation of urban forms, integrating both street networks and plot systems with buildings within a unified topological structure.

The heterogeneous graph constructed by `morphological_graph()` could be a foundational data structure for policy evaluation at the neighbourhood scale. For example, the evaluation of the 15-Minute City requires an understanding of proximity enabled by the street network, as well as the concentration and mixture of urban functions (Moreno et al., 2021). In a morphological graph, both of these types of information can be accommodated through the topological structure and node attributes, and can be applied to analytical tasks such as the measurement of urban form and function, and the identification of typologies through clustering.

3.1.2. Transportation

City2Graph's transportation module focuses on the processing of GTFS data and the creation of transportation network graphs. The core workflow involves the `load_gtfs()` function, which parses GTFS zip files and enriches stop and route data with spatial geometries, and the `travel_summary_graph()` function, which constructs a graph summarising trips of transports from the processed GTFS data. In travel summary graphs, nodes represent stops, while edges indicate scheduled transit operations, weighted by service frequency and average travel time under a specified calendar term and/or time range within a day. An example of subways in New York City is shown in Fig. 4(a).

By aggregating individual scheduled trips into weighted stop-to-stop connections, this module converts the schedule-oriented GTFS format into a form directly usable for transportation network analysis and traffic flow prediction. Moreover, combining them from multiple modes (e.g., bus and railway) or from the street network constructed by the morphology module (Section 3.1.1) can extend the tasks to multimodal scenarios.

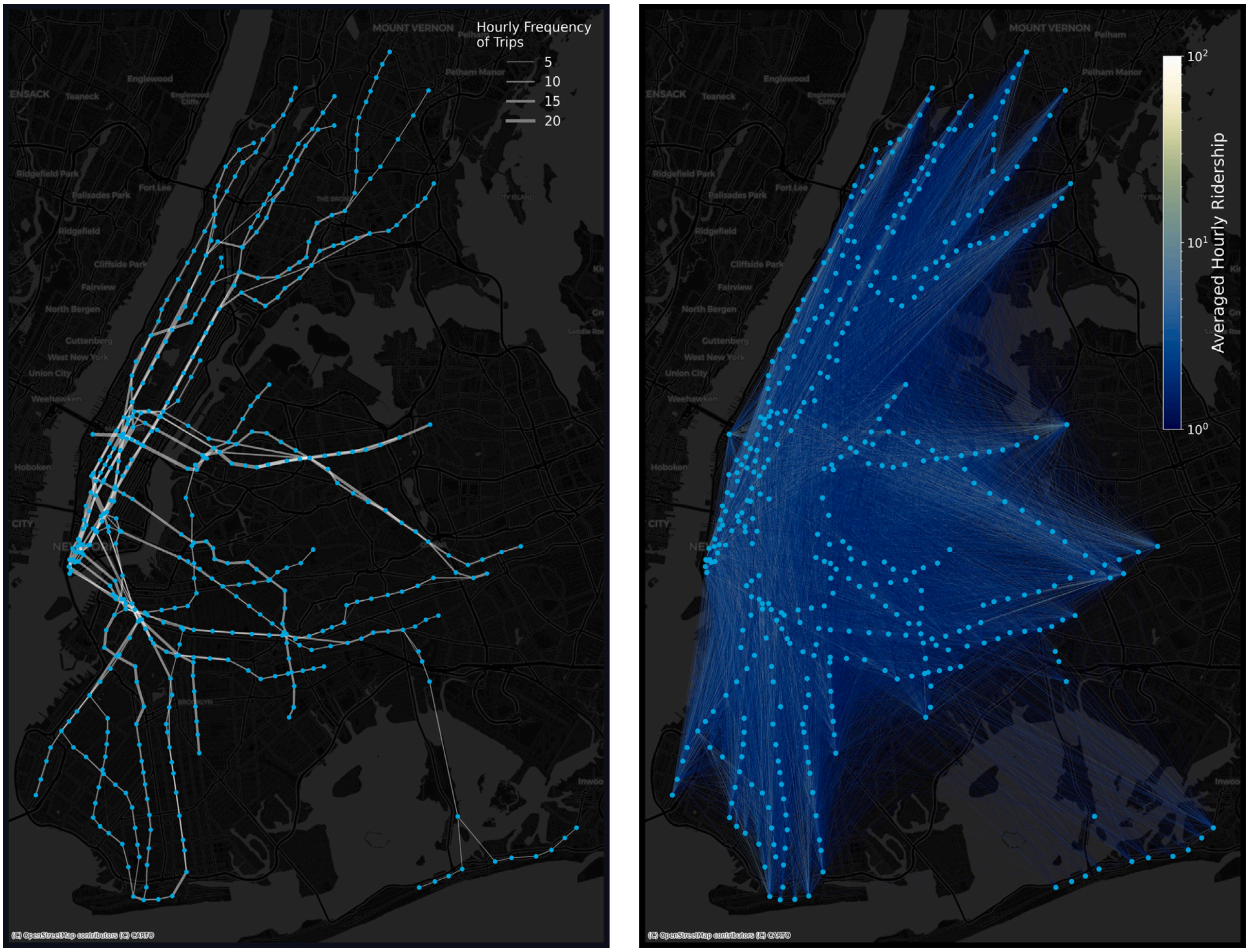
3.1.3. Mobility

The mobility module deals with mobility data (e.g., subway ridership, bike-sharing transactions, and migration patterns) in the form of origin–destination (OD) matrices. The `od_matrix_to_graph()` function converts OD matrices, either as edge lists or an adjacency matrix, into the standardised graph structure of City2Graph, where nodes are specified as zones represented by polygons or points, and edges represent flows between them. With the same extent as Fig. 4(a), the OD matrix of subway ridership in New York City is shown in Fig. 4(b). As nodes correspond to identifiable zones or stations, users can directly attach node attributes on GeoPandas (e.g., zonal population, employment, or land-use composition), providing the basis for tasks such as edge prediction of mobility flows.

3.1.4. Proximity

The proximity module provides functions that construct edges between spatial objects whose transportation, mobility, or functional relations are not directly observed, by flexibly designing what counts as geometric closeness. Throughout, the library supports several distance metrics, including Manhattan (L1), Euclidean (L2), and network-based distances derived from provided linestring geometries, as shown in Fig. 5.

For points, multiple construction methods are offered, each reflecting a different definition of proximity. `knn_graph()` connects each point to its k closest neighbours, yielding a graph based on relative proximity by the order of distance to surroundings. By contrast, `fixed_radius_graph()` (Bentley et al., 1977) connects points within a specified threshold, which is suitable for situations where absolute distance has a physical meaning, such as walking catchments or sensor coverage. While those two algorithms require parameters, non-parametric algorithms are also provided, such as `delaunay_graph()` (Lee & Schachter, 1980), `gabriel_graph()`



(a) Travel Summary Graph

(b) OD Matrix

Fig. 4. Visualisation of (a) travel summary graph and (b) OD matrix of subway in New York City. Both figures depict a snapshot from 06:00 to 10:00 on Nov. 3, 2025. (a) Nodes represent transit stops, while edges denote scheduled trips with thickness indicating the hourly frequency of trips. (b) Graph derived from OD matrix data, where edge colour signifies the averaged hourly ridership on a logarithmic scale. Both datasets were obtained from The Metropolitan Transportation Authority: GTFS from https://www.mta.info/developers (accessed on Nov. 3, 2025) and Subway Origin–Destination Ridership Estimates from https://data.ny.gov/Transportation/MTA-Subway-Origin-Destination-Ridership-Estimate-B/y2qv-fytt/about_data.

(Gabriel & Sokal, 1969), `relative_neighborhood_graph()`, and `euclidean_minimum_spanning_tree()`, alongside the distance-decay `waxman_graph()`. For polygons, `contiguity_graph()` constructs Queen or Rook adjacency based on shared boundaries, wrapping libpysal (Rey & Anselin, 2010; Rey et al., 2022). These graphs have potential for a range of downstream analyses. The choice among these methods is not a neutral preprocessing step but encodes a hypothesis about which spatial relationship is analytically relevant. With POIs as nodes and their proximity as edges, community detection or centrality analysis could be applied for identifying co-located amenity clusters and local hubs. If GNNs are applied to administrative boundaries with their contiguity, it enables geodemographic classification that explicitly incorporates neighbourhood effects (De Sabbata & Liu, 2023).

To facilitate heterogeneous graph constructions, this module provides dedicated inter-layer connection strategies. `bridge_nodes()` generates directed proximal edges between distinct semantic layers, extending *k*NN and fixed-radius logic to couple heterogeneous entities via geometric or network distances, as seen in Fig. 6(a). Complementarily, `group_nodes()` encodes hierarchical spatial dependencies, linking point geometries to enclosing polygons through spatial predicates.

Between polygons, spatial adjacency is managed via `contiguity_graph()`, utilising either Queen or Rook adjacency as a wrapper for libpysal (Rey & Anselin, 2010; Rey et al., 2022). Although this function is designed for a single layer of polygons, its combination with the other proximity functions can capture complex spatial structures as heterogeneous graphs. For instance, Fig. 6(b) demonstrates a composite graph structure where `group_nodes()` links points to polygons while `contiguity_graph()` simultaneously establishes the neighbourhood structure between those polygons.

3.1.5. *Metapath*

In heterogeneous graphs, a metapath can be identified as a semantic relation between two nodes established through a sequence of intermediate steps with multiple edge types (Sun & Han, 2013). As illustrated in Fig. 7, while standard spatial graphs encode direct physical topology, metapaths capture higher-order proximity. For instance, a traversal across multiple street segments can be abstracted into a single functional connection if the nodes are reachable within a defined threshold. Metapath projection facilitates message-passing of heterogeneous GNNs between spatially distant yet functionally related nodes, enabling models to capture complex patterns such as accessibility and land use complementarity.

City2Graph operationalises this concept through two functions that identify these semantic relations as explicit edges. The `add_metapaths()` function constructs edges by traversing specified relation sequences (Fig. 8(a)), aggregating attributes such as travel time and preserving route geometries for visualisation. Complementarily, `add_me`

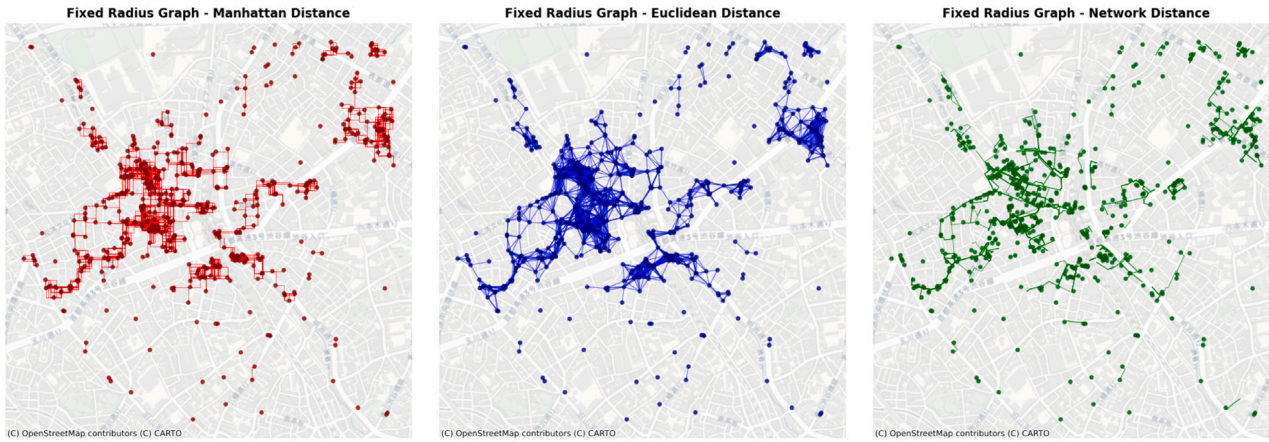


Fig. 5. Visualisation of fixed-radius graph. Nodes are connected if the distance is within a threshold of 100 m. For nodes, restaurant POIs are extracted from OSM in Shibuya, Tokyo.

tapaths_by_weight() generates threshold-based projections (Fig. 8(b)), employing Dijkstra's algorithm to establish direct connectivity between nodes separated by a specific cumulative cost, such as a 15-minute walking catchment.

This contrasts with approaches that encode spatial relations as text (e.g., "Place A is walkable from station B") and learn node embeddings through skip-gram models, such as Place2vec for POIs (Zhai et al., 2019). Whereas such embeddings capture co-occurrence only implicitly, metapaths preserve these relations as explicit typed edges with weights such as travel time, so that different semantic assumptions can be specified directly (e.g., 15-minute walking vs. multimodal accessibility).

3.2. Graph conversion

Rather than introducing a custom graph class, the library deliberately reuses the GeoDataFrame tuples described in Section 3.1 as its canonical data structure for inputs. All functions internally employ converters that inherit from a shared abstract class (BaseGraphConverter) that automatically dispatches to homogeneous or heterogeneous graphs depending on whether the input is GeoDataFrames or typed dictionaries. Each conversion attaches a metadata object to the resulting graph that records the coordinate reference system, node and edge type registries, original index names, a lookup table that maps each original index value (e.g., a census area code) to the sequential integer node identifier used by PyTorch Geometric, and the serialised geometries in Well-Known Binary (WKB) format. This metadata is what enables the inverse conversion to reconstruct GeoDataFrames that are structurally identical to the originals, including their spatial reference and index hierarchy.

The conversion pipeline relies on vectorised operations (NumPy array stacking and Pandas index mapping) rather than element-wise Python loops, which keeps execution time manageable for at least city-scale graphs. As a reference, the case study of Liverpool in Section 4, comprising 1624 nodes and 102 450 undirected edges (204 900 directed entries in edge_index) across three relation types, completed the full conversion process without performance bottlenecks on a laptop environment (0.620 s by gdf_to_pyg() on an Apple M2 16 GB). Optionally, users can reduce memory overhead by disabling geometry serialisation if only node positions are needed for reconstruction.

3.2.1. NetworkX

gdf_to_nx() and nx_to_gdf() enable lossless bidirectional conversion between GeoDataFrames and NetworkX (Hagberg et al., 2008) for homogeneous and heterogeneous graphs, utilising GraphMetadata to preserve indices and geometries. To support heterogeneous graphs in NetworkX, nodes store their types and offset-based identifiers as attributes, while edges are labelled with type triplets. Additionally, integration with rustworkx (Treinish et al., 2022) is supported via nx_to_rx() and rx_to_nx(), providing access to a Rust-backed graph engine for operations where execution speed is critical. With this operation, measures of network science including centralities can be computed and seamlessly appended back as node attributes to be used for spatial analysis with visualisation, or GNNs.

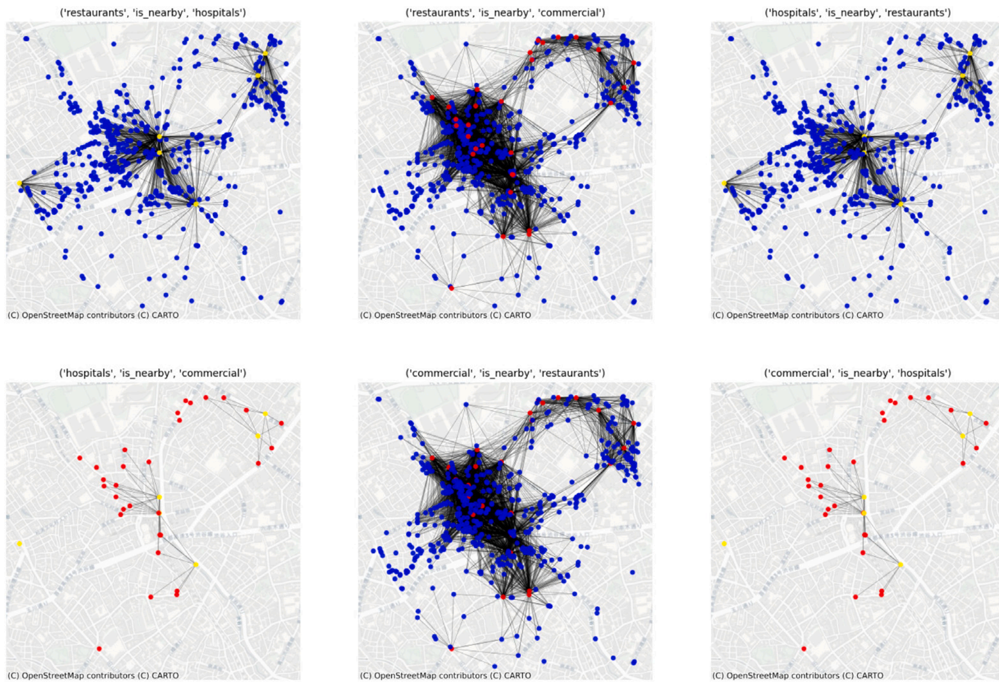
3.2.2. PyTorch Geometric

City2Graph implements conversion to and from PyTorch Geometric for the model training of GNNs. The gdf_to_pyg() function serialises constructed graphs into Data (homogeneous) or HeteroData (heterogeneous) objects. During this step, user-specified columns are extracted as node feature, node label, and edge feature tensors, while a mapping between original GeoDataFrame indices and the sequential integer indices required by PyTorch Geometric is recorded in the metadata. Tensor placement on CPU or GPU can be specified at conversion time. The reverse function (pyg_to_gdf()) uses the stored metadata to map tensors, including any new attributes added during training, such as learned embeddings or attention weights, back into GeoDataFrames with their original index, geometry, and coordinate reference system. Conversions are also available between NetworkX and PyTorch Geometric by nx_to_pyg() and pyg_to_nx().

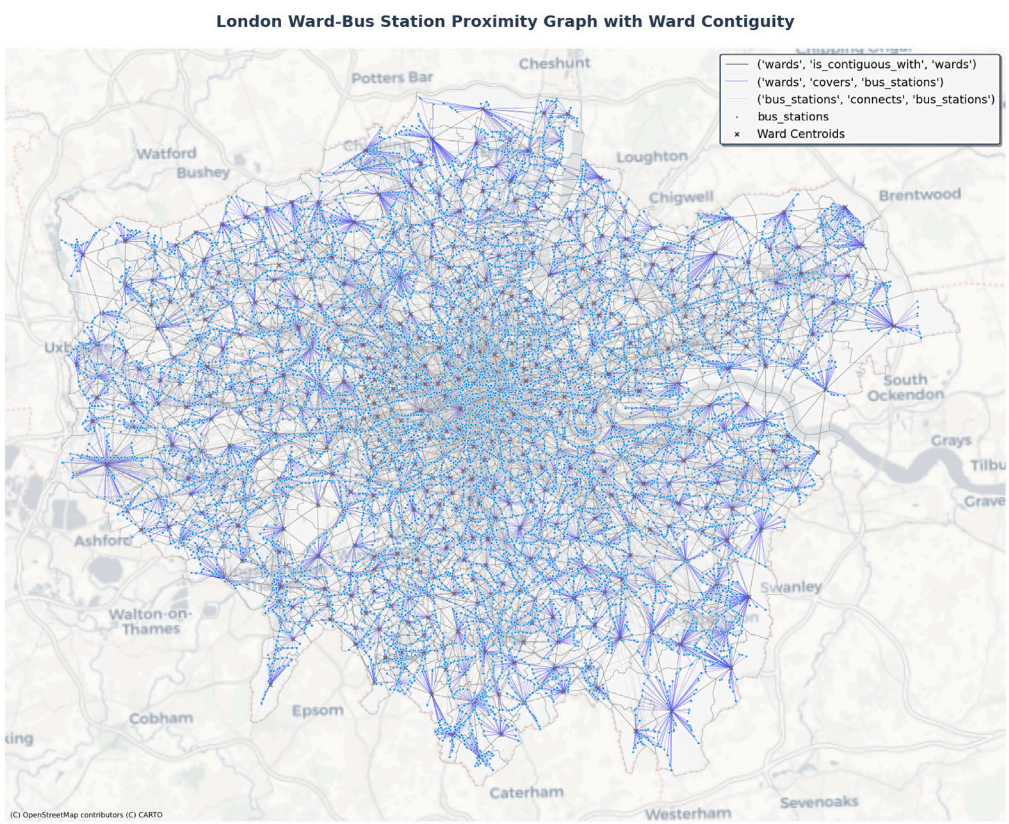
3.3. Graph interpretation

City2Graph supports the model interpretation of GNN outcomes through integrated functions for visualisation and spatial analysis. The plot_graph() function addresses the complexity of rendering heterogeneous structures by providing a unified visualisation interface for NetworkX objects and GeoDataFrames, automatically handling spatial alignment. It enables the simultaneous visualisation of diverse nodes and edges with customisable visual attributes (e.g., colour, transparency) mapped directly from feature columns. Additionally, automated subplot layouts for semantic layers aid in the qualitative assessment of graph topology and feature distributions.

The create_isochrone() function supports accessibility analysis and the definition of local neighbourhoods by computing service areas from a centre point within a user-defined cost threshold using Dijkstra's algorithm. To represent spatial extents, it offers multiple geometric construction methods, including convex hulls, concave hulls based on k-nearest neighbours (Moreira & Santos, 2007) or alpha shapes (Gillies et al., 2025), and edge buffering. When a heterogeneous graph is provided, users can specify edge types to ignore during polygon generation and use them only for distance calculation. In multimodal accessibility analyses that combine streets with a transit system, stop-to-stop edges can connect distant areas, but they should be excluded from the polygons because walking is only possible from each stop.



(a) Bridging Nodes



(b) Grouping Nodes on Contiguity Graph

Fig. 6. Visualisation of node manipulation tools for heterogeneous and homogeneous graph construction. (a) Bridging nodes: directed proximity edges are generated between different layers of nodes. POIs of restaurants are extracted from OSM in Shibuya, Tokyo. (b) Grouping nodes and Contiguity: spatial relationships are generated between polygon and point geometries based on spatial predicates, while polygons are connected by queen-based contiguity. Lower Layer Super Output Areas as wards are from the Office for National Statistics (https://geoportal.statistics.gov.uk/datasets/6beafcfd9b9c4c9993a06b6b199d7e6d_0), and bus stops are from Transport for London (https://data.bus-data.dft.gov.uk/timetable/download/gtfs-file/london/).

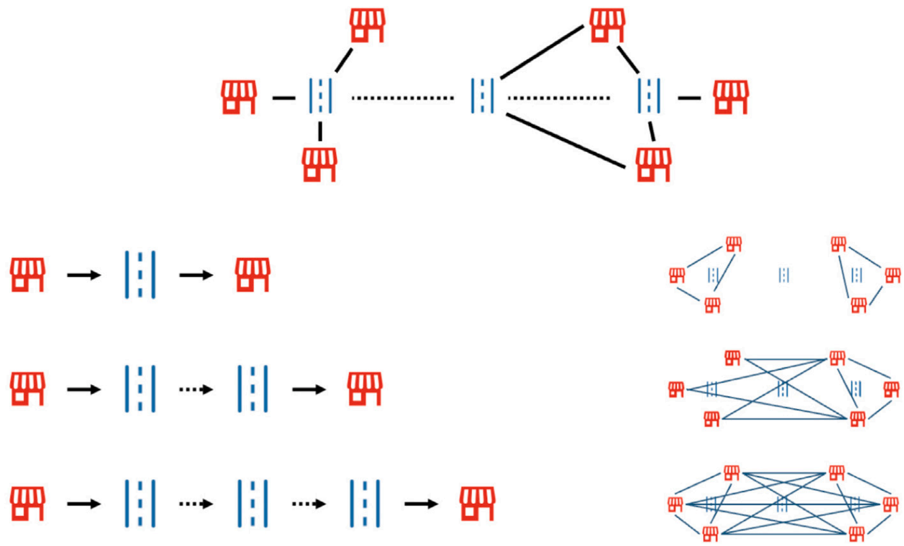


Fig. 7. Conceptual illustration of metapaths. The heterogeneous graph (top) connects entities through intermediate nodes (e.g., amenities connected via street segments), whereas the metapath projection (bottom) establishes direct semantic edges between them, capturing higher-order proximity such as accessibility.

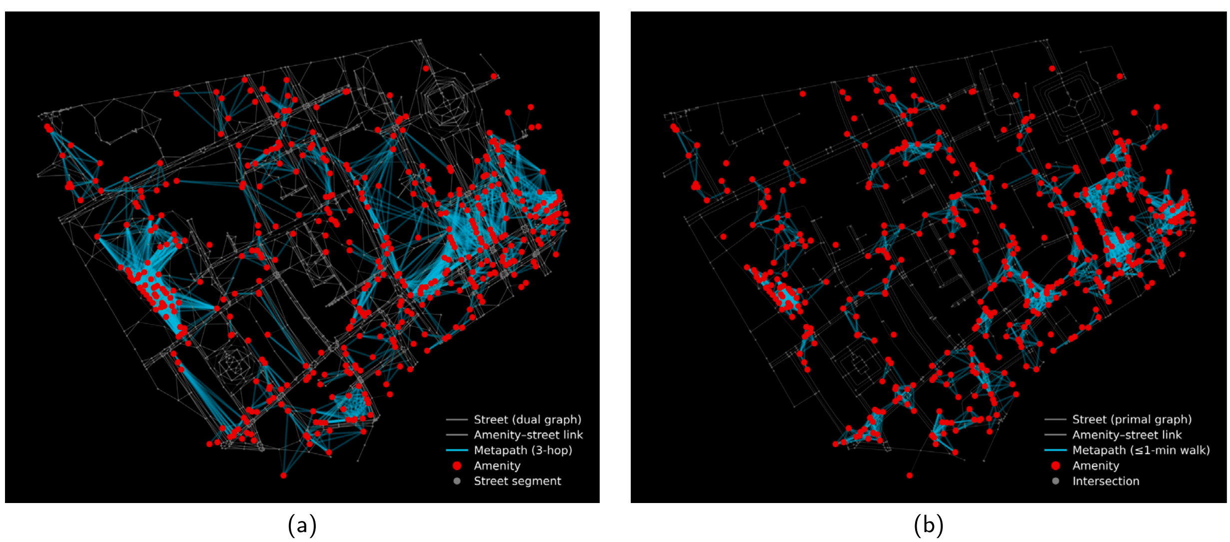


Fig. 8. Visualisation of metapath construction. (a) Composite relations generated by `add_metapaths()`, connecting amenities separated by 3 hops on the dual graph of streets (node: segment, edge: intersection). (b) Distance-based connectivity generated by `add_metapaths_by_weight()`, linking amenities within a 1-minute walking distance (80 m) on the primal graph of streets (node: intersection, edge: segment). Amenity nodes were bridged to the nearest neighbour of street graph nodes by `bridge_nodes()`. Data were extracted from OSM via OSMnx for Soho, London. Amenities included POIs tagged as "cafe", "restaurant", "pub", "bar", "museum", "theatre", or "cinema".

3.4. Other functionalities

Although Overture Maps provides high-quality building footprints and transportation networks, its complex nested schema and columnar Parquet format present barriers to researchers unfamiliar with big data infrastructure (Ballantyne & Berragan, 2024). City2Graph addresses these challenges through the `load_overture_data()` function, which retrieves multiple data types for any geographic area and returns GeoDataFrame objects, and the `process_overture_segments()` function, which preprocesses the topology of street segments by splitting at connectors, clustering endpoints to resolve coordinate discrepancies, and parsing metadata to identify impassable road portions.

4. Case study

This case study illustrates the end-to-end efficacy of City2Graph for applying heterogeneous GNNs to urban systems, from data collection and preprocessing, through heterogeneous graph construction and PyTorch Geometric conversion, to spatial interpretation of model outputs. We learn unsupervised low-dimensional node embeddings of census areas that integrate attributes of urban functions with contiguity and accessibility relations. Clusters are obtained as groups of census units that share similarities in terms of the composition of urban functions (POIs and land use), as well as their connectivities between each other. Fig. 9 provides an overview of this workflow, illustrating how the whole pipeline can be supported by City2Graph. Reproducible notebooks demonstrating this workflow are available at https://github.com/c2g-dev/city2graph-case-study.

4.1. Study area and spatial unit

This case study highlights Liverpool, United Kingdom. Following population decline in the late twentieth century, the city has experienced uneven recovery, characterised by rapid central growth in the

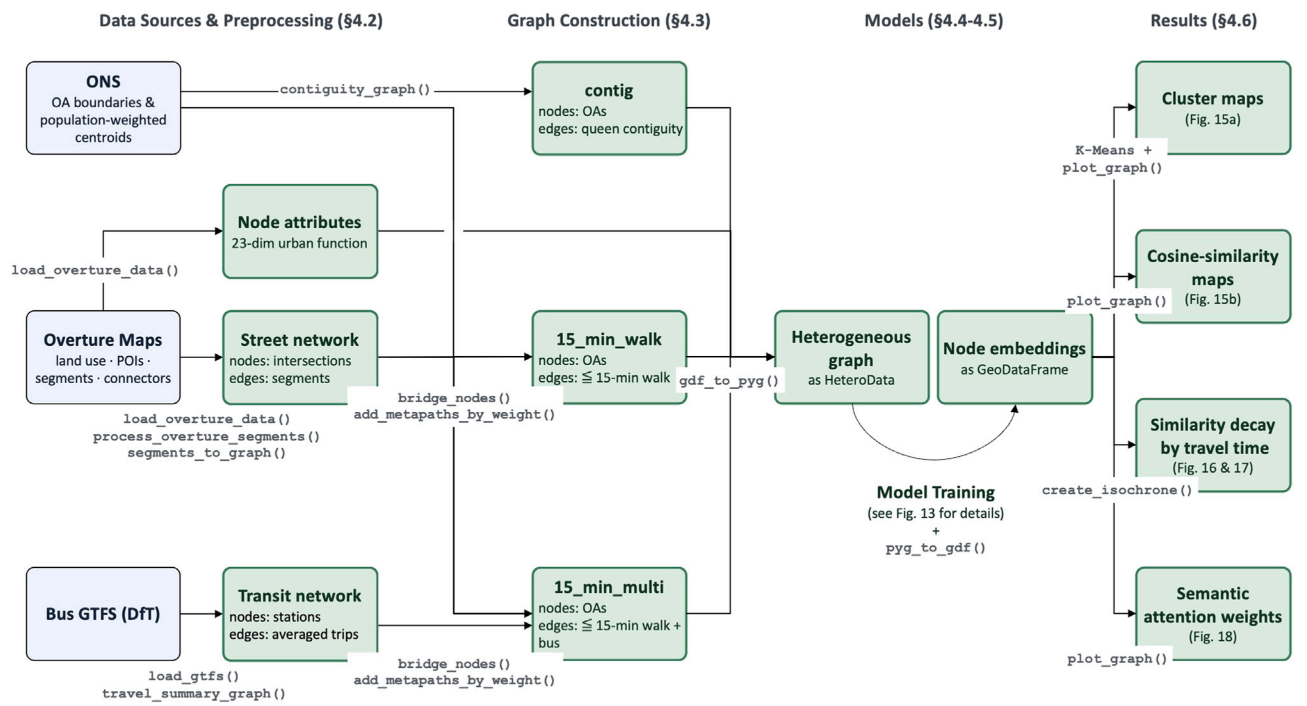


Fig. 9. Overview of the Liverpool case study workflow, annotated with the City2Graph function applied at each stage. Data sources and preprocessing (Section 4.2), graph construction of contiguity and accessibility relations (Section 4.3), heterogeneous-graph conversion and model training (Section 4.5), and spatial interpretation of outputs (Section 4.6). ONS, OA, and DfT stand for Office for National Statistics, Output Area, and Department for Transport, respectively.

north-west, driven by regeneration projects such as the 'Liverpool ONE' shopping centre and waterfront redevelopment (Sykes et al., 2013), alongside stagnation in outer areas (Couch & Fowles, 2019). These two tendencies have fostered spatial inequalities in access to public and private services (Calafiore et al., 2022). Liverpool is therefore a suitable case for identifying clusters of neighbourhoods from the combined patterns of urban functions and accessibility. Output Areas (OAs) were employed as the spatial unit ($N = 1624$). OAs are the smallest standard census unit in the UK and are constructed to contain between 40 and 250 households and a resident population of 100 to 625 persons (Office for National Statistics, 2021). This design ensures broadly comparable population sizes, supporting neighbourhood-scale inference while retaining interpretability for planning and policy contexts (Lloyd, 2016).

4.2. Data preprocessing

For the case study, three data sources were employed. First, OA geometries and population-weighted centroids were retrieved from the Office for National Statistics.[1] Second, urban function was represented using POIs and land use from Overture Maps, as shown in Fig. 10. These layers were downloaded reproducibly for a Liverpool bounding box using `load_overture_data()` (release: 2025-12-17.0, accessed on Jan. 17, 2026) and aggregated to OA-level indicators, producing a 23-dimensional vector per OA. Third, the transportation network was derived from (i) street segments and connectors from the same Overture Maps release and (ii) a bus GTFS feed of the North West obtained from the Department for Transport.[2] City2Graph was used to split Overture Maps segments by connectors, indicators of routing, using `process_overture_segments()`, then convert them into nodes and edges via `segments_to_graph()`. The GTFS feed was parsed with `load_gtfs()` and summarised into a weighted stop-to-stop network using `travel_summary_graph()`, where nodes represent bus stops and edges represent trips between them, with the edge attribute of average travel time in seconds over the planned operation period from Dec. 10, 2025 to Sep. 10, 2026.

4.3. Graph construction

4.3.1. Contiguity

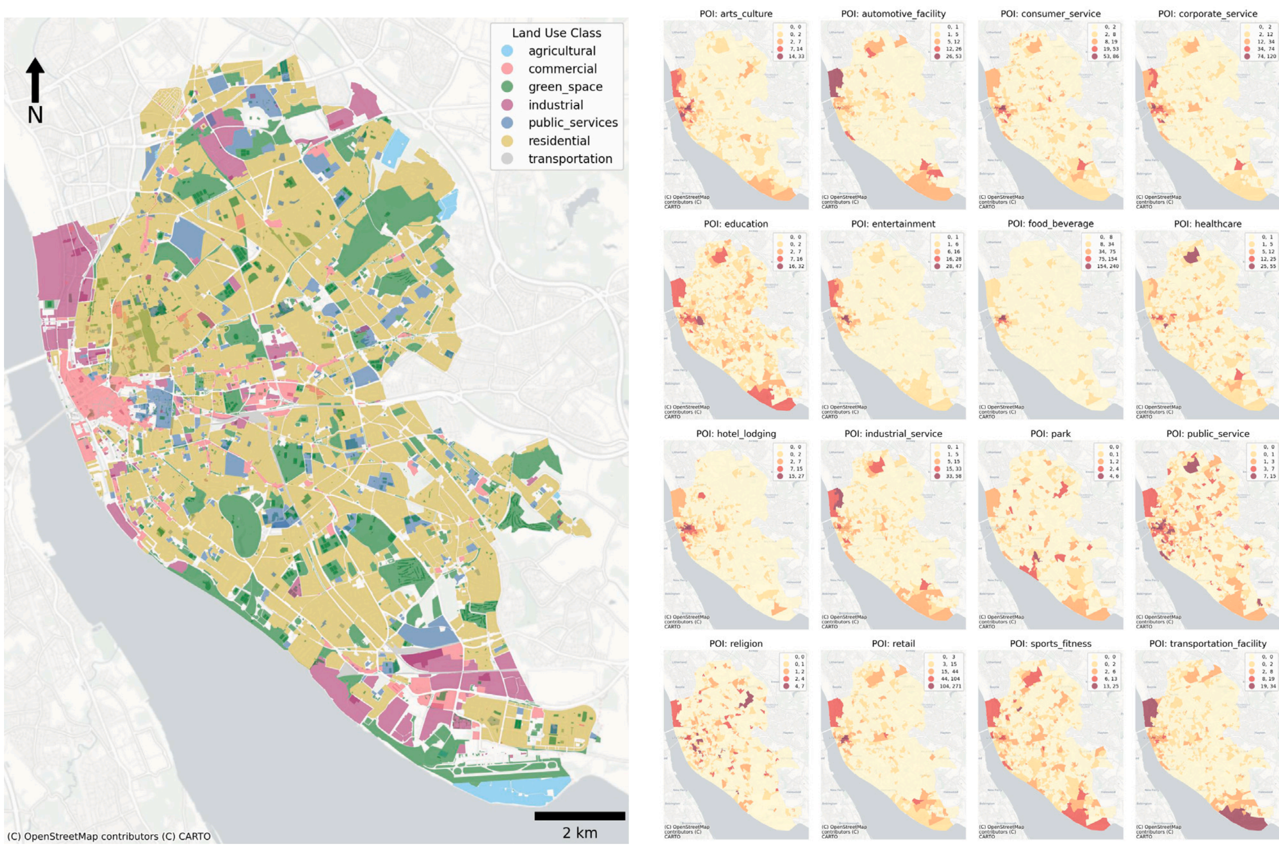
For comparison, a homogeneous graph of OAs was constructed, defined by Queen contiguity. City2Graph generated this layer using `contiguity_graph()`, representing OAs as point nodes at population-weighted centroids. The resulting node and edge GeoDataFrames were converted into a PyTorch Geometric `Data` object using `gdf_to_pyg()`, preserving identifiers and coordinate reference system for subsequent inverse mapping. Fig. 11 shows the OA polygons and centroids in panel (a), and the resulting contiguity graph in panel (b).

4.3.2. Multimodal accessibility as metapaths

In addition to contiguity, accessibility-based OA–OA relations were constructed across the street and bus transit networks to capture the multimodal structure of the city. To integrate multiple spatial layers, the `bridge_nodes()` function was employed to link nodes of different types to their nearest counterparts. For this operation, connections were established between OA centroids and street connectors, and between street connectors and bus stops. With the `add_metapaths_by_weight()` function, multi-hop paths were projected into direct edges between OAs, establishing connectivity for pairs within a 15-minute (900 s) travel time threshold. Two accessibility-based relations were defined as shown in Fig. 12, with walk-based connectivity in panel (a) and transit-assisted connectivity in panel (b). The *15_min_walk* relation captures local walk-based accessibility, linking OAs reachable via the street network. The *15_min_multi* relation represents transit-enabled

[1] Data sources: OA boundaries (https://geoportal.statistics.gov.uk/datasets/6beafcfd9b9c4c9993a06b6b199d7e6d_0) and population-weighted centroids (https://geoportal.statistics.gov.uk/datasets/ons::output-areas-december-2021-ew-population-weighted-centroids-v3).

[2] Data source: https://findtransportdata.dft.gov.uk/dataset/bus-open-data—download-all-timetable-data–18335fb19c4. Accessed on Dec. 10, 2025.



(a) Land Use (b) POIs

Fig. 10. Visualisation of urban function data in Liverpool: (a) Land use classification and (b) counts of POIs by OA. Data retrieved from Overture Maps (accessed on Jan. 17, 2026).

accessibility, connecting OAs via the integrated walking and bus transit network. Note that this relation explicitly excluded pairs that were already connected by *15_min_walk* to ensure the structural distinction provided by public transport. Finally, these relations were synthesised into PyTorch Geometric `HeteroData` objects via `gdf_to_pyg()`.

4.4. Problem formulation

The clustering task in this case study is formulated as unsupervised representation learning. The objective is to learn low-dimensional node embeddings that jointly encode (i) urban function attributes and (ii) multiple relations that capture spatial contiguity and accessibility. The resulting embeddings are then used as inputs to a clustering algorithm to derive functional clusters.

Let $\mathcal{V} = \{1, \ldots, N\}$ denote the set of OAs ($N = 1624$). Each OA i has a feature vector $\mathbf{x}_i \in \mathbb{R}^F$ ($F = 23$), producing $\mathbf{X} \in \mathbb{R}^{N \times F}$. Features were log-transformed and standardised

$$\tilde{x}_{i,f} = \frac{\log(1 + x_{i,f}) - \mu_f}{\sigma_f} \tag{1}$$

where μ_f and σ_f are the mean and standard deviation over $\log(1+x_{i,f})$, respectively. Stacking the standardised vectors yields $\tilde{\mathbf{X}} \in \mathbb{R}^{N \times F}$.

Connectivity was encoded by a set of relations, $\mathcal{M}$. Each relation, $m \in \mathcal{M}$, defines a set of edges $\mathcal{E}^{(m)} \subseteq \mathcal{V} \times \mathcal{V}$ with an associated travel-time attribute, $\tau_{ij}^{(m)}$, in seconds on each edge. All relations were treated as undirected and included (i) Queen contiguity (m = *contig*), (ii) 15-minute walking accessibility (m = *15_min_walk*), and (iii) 15-minute transit-assisted accessibility (m = *15_min_multi*). Contiguity edges were assigned a pseudo travel time $\tau_{ij}^{(contig)} = d_{\text{euc}}(i,j)/v_{\text{walk}}$, where $d_{\text{euc}}(i,j) \in \mathbb{R}_{\geq 0}$ denotes the Euclidean distance between population-weighted centroids and $v_{\text{walk}} = 4.8$ km/h, adhering to the standard walking speed defined by the Department for Transport,[3] as shown in Fig. 11. Edge travel times were also transformed as

$$\tilde{\tau}_{ij}^{(m)} = \frac{\log(1 + \tau_{ij}^{(m)}) - \mu_\tau^{(m)}}{\sigma_\tau^{(m)}} \tag{2}$$

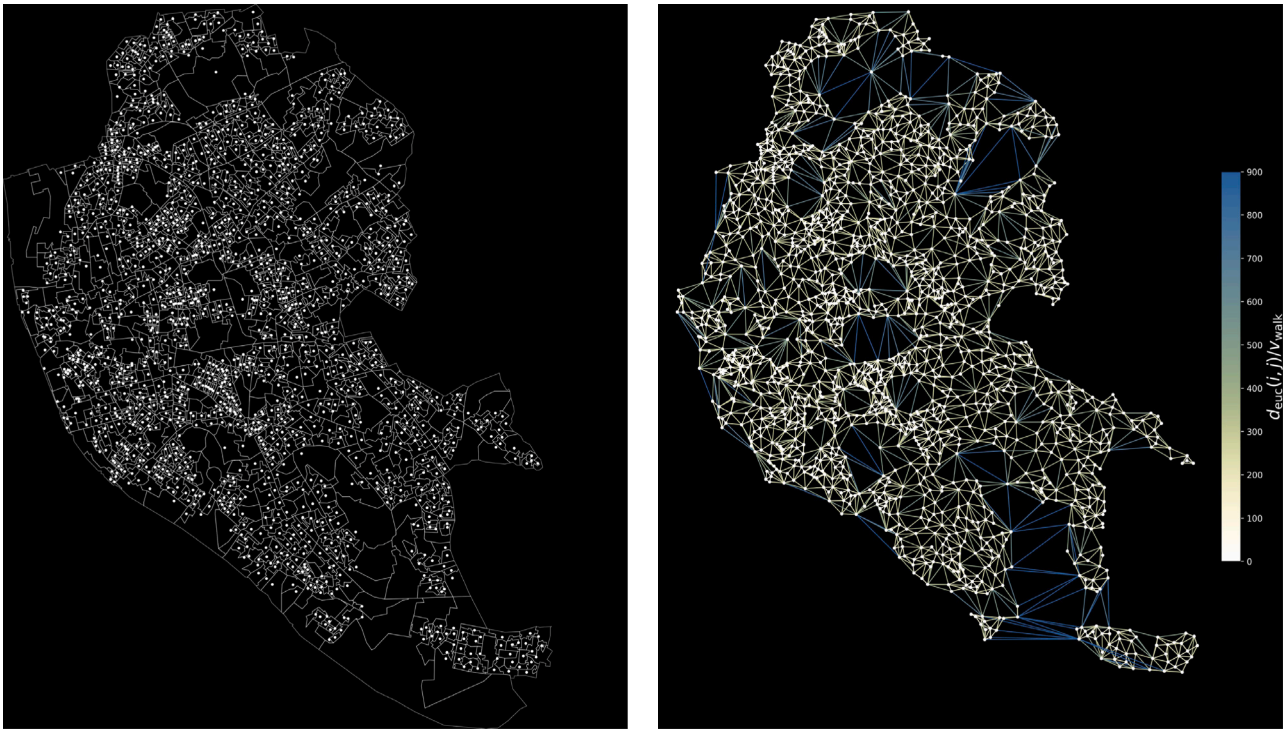
where $\mu_\tau^{(m)}$ and $\sigma_\tau^{(m)}$ are computed over $\log(1+\tau_{ij}^{(m)})$ for edges in relation m. For each relation m, the vector of standardised travel times over all edges is denoted $\tilde{\boldsymbol{\tau}}^{(m)} \in \mathbb{R}^{|\mathcal{E}^{(m)}|}$.

4.5. Models

Four Graph Autoencoder (GAE) architectures with attention layers are compared (Kipf & Welling, 2016; Salehi & Davulcu, 2019; Wang et al., 2023). Fig. 13 illustrates the overall architecture of the case study. In this context, a GAE learns a compact embedding for each OA such that (i) OAs with similar urban functions have similar embeddings and (ii) OAs that are connected under the chosen relations can be recovered from the embeddings. This is achieved by an encoder that performs message-passing on the graph, and decoders that reconstruct node attributes and relation-specific connectivity.

Attention is adopted for two methodological reasons and one interpretative benefit. First, the influence of neighbouring nodes is non-uniform; even within the same travel-time threshold, some connections are more informative of shared function than others, and attention learns these neighbour contributions rather than imposing them. Second, as travel time represents a cost of interaction, integrating this

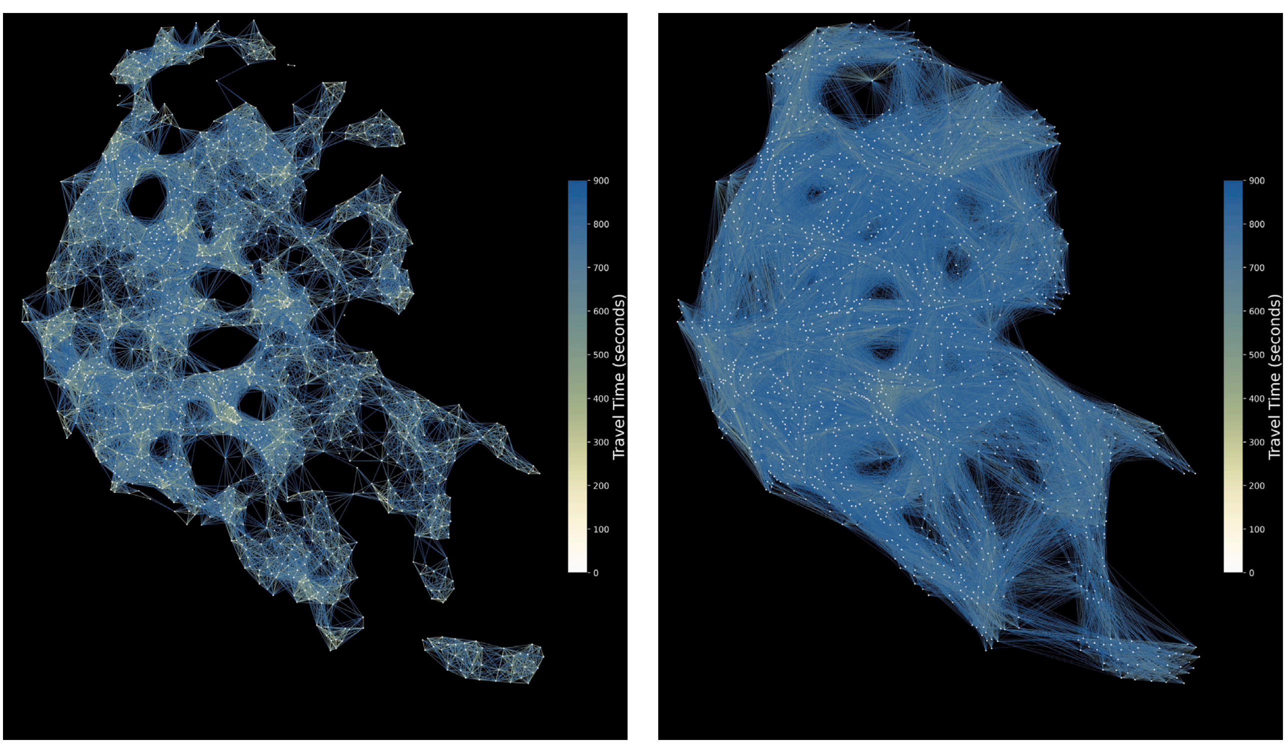
[3] https://www.gov.uk/government/publications/journey-time-statistics-guidance/journey-time-statistics-notes-and-definitions-2019



(a) OAs and population-weighted centroids

(b) Queen-contiguity graph

Fig. 11. Visualisation of the OA layer and contiguity graph in Liverpool: (a) Output Area (OA) polygons with their population-weighted centroids, used as graph nodes; and (b) Queen-based contiguity edges between centroids. For edge (i, j), $d_{\text{euc}}(i, j)/v_{\text{walk}}$ denotes the spatial weight computed from Euclidean distance and a walking speed of 4.8 km/h.



(a) *15_ min_ walk* metapaths

(b) *15_ min_ multi* metapaths

Fig. 12. Spatial distribution of accessibility-based OA–OA relations as metapaths in Liverpool: (a) the *15_min_walk* relation connects OAs within 15 min walking distance; and (b) the *15_min_multi* relation connects OAs accessible within 15 min via bus-based public transport, excluding pairs already connected by *15_min_walk*. Edge colours indicate travel time in seconds.

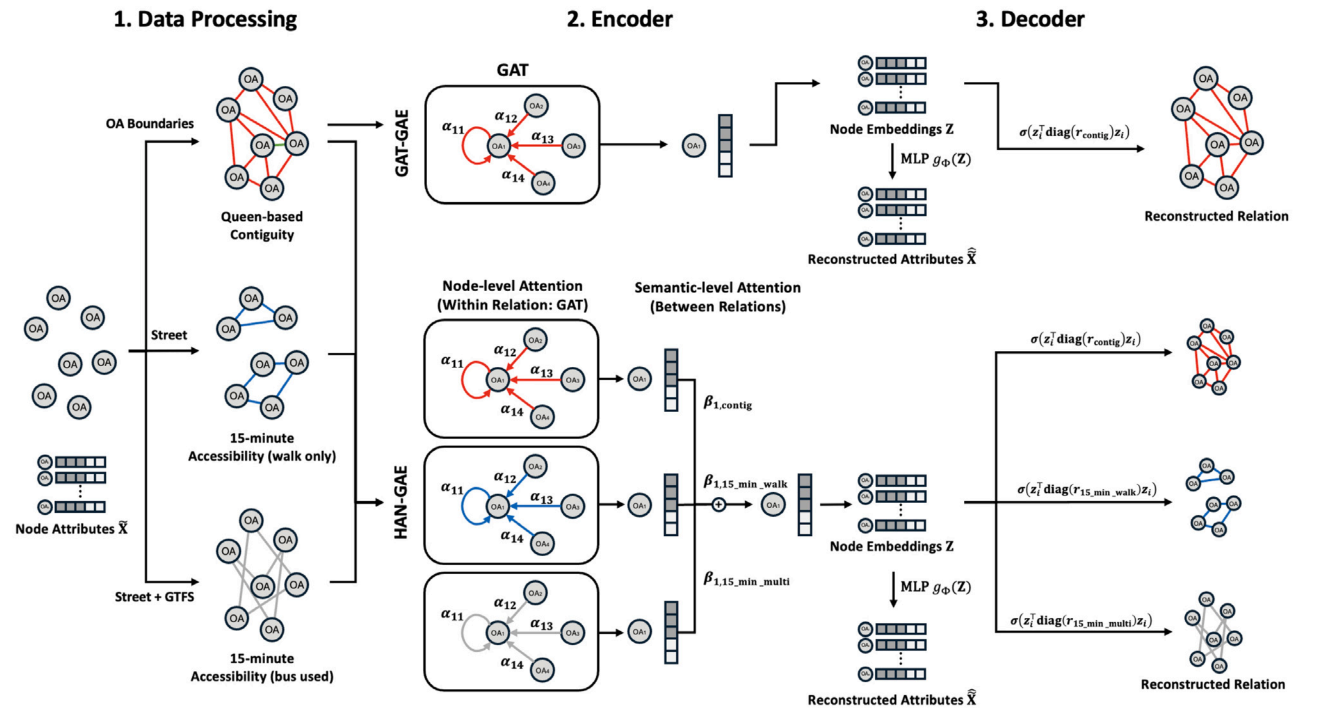


Fig. 13. Architecture of the four Graph Autoencoder models compared in the case study: data processing of node attributes and the three relations (left), edge-aware GAT/HAN encoders with node- and semantic-level attention (centre), and attribute/structure decoders (right).

attribute into attention conditions message-passing on $\tilde{\tau}_{ij}^{(m)}$, avoiding the unrealistic assumption of uniform edge influence. Third, in heterogeneous settings, semantic attention between relations produces coefficients that indicate how strongly each relation contributes to each OA's embedding, which enhances interpretability through relation-wise attribution (see Section 4.6.3).

Each model has an edge-aware encoder f_θ that maps scaled attributes and relation-specific edge sets (with travel-time attributes) to d-dimensional embeddings $\mathbf{Z} \in \mathbb{R}^{N\times d}$ such that

$$\mathbf{Z} = f_\theta\left(\tilde{\mathbf{X}}, \{(\mathcal{E}^{(m)}, \tilde{\tau}^{(m)})\}_{m\in\mathcal{M}}\right). \tag{3}$$

Subsequently, node attributes and relation-specific structures are reconstructed from these embeddings by decoders g_ψ and d_ϕ, producing

$$\hat{\tilde{\mathbf{X}}} = g_\psi(\mathbf{Z}) \in \mathbb{R}^{N\times F}, \quad \hat{\mathbf{A}}^{(m)} = d_\phi(\mathbf{Z}, m) \in [0,1]^{N\times N} \tag{4}$$

for each relation $m \in \mathcal{M}$, respectively. Note that $\hat{\mathbf{A}}^{(m)} = \left[p_m(i,j)\right]$ where $p_m(i,j) \in [0,1]$ is the predicted probability of an edge between nodes i and j under relation m.

The homogeneous model (Model 1) uses a Graph Attention Network (GAT) encoder (Veličković et al., 2018) with $\mathcal{M} = \{contig\}$. Heterogeneous configurations (Models 2–4) use a Heterogeneous Graph Attention Network (HAN) encoder (Wang et al., 2019), adapted to condition attention on edge travel times, which computes relation-specific embeddings and then combines them via semantic attention between relations. The resulting semantic attention weights $\beta_{i,m}^{(l)}$ at each layer can be interpreted as the relative contribution of relation m to the embedding of OA i.

4.5.1. Encoders

In each encoder, node attributes are aggregated from neighbours and updated by GAT layers, denoted as GAT(·). In this study, the aggregation is edge-aware; the contribution of a neighbour is conditioned not only on node attributes but also on the scaled travel time $\tilde{\tau}_{ij}^{(m)}$. We employ multi-head attention with $H = 4$ heads to stabilise the learning process (Sun et al., 2023), concatenating the weighted contributions of each head. The full mathematical formulation of GAT(·) is provided in Appendix B.1.

Model 1 (GAT-GAE). Model 1 employs a single-relation encoder over the contiguity graph ($\mathcal{M} = \{contig\}$). The node representation at layer $l+1$ is computed as:

$$\mathbf{h}_i^{(l+1)} = \mathrm{GAT}\left(\mathbf{h}_i^{(l)}, \{(\mathcal{N}_{contig}(i), \tilde{\boldsymbol{\tau}}^{(contig)})\}\right) \tag{5}$$

where $\mathbf{h}_i^{(0)} = \tilde{\mathbf{x}}_i$ and $\mathcal{N}_{contig}(i)$ denotes the set of neighbours.

Models 2–4 (HAN-GAE). Models 2–4 employ a HAN-based encoder with multiple relations. First, node attributes are updated separately for each relation $m \in \mathcal{M}$ by the edge-aware GAT mechanism:

$$\mathbf{h}_i^{(l+1,m)} = \mathrm{GAT}\left(\mathbf{h}_i^{(l)}, \{(\mathcal{N}_m(i), \tilde{\boldsymbol{\tau}}^{(m)})\}\right) \tag{6}$$

These relation-specific embeddings are then combined via semantic attention to form a unified representation $\bar{\mathbf{h}}_i^{(l+1)}$. A learnable attention vector $\mathbf{q}^{(l)}$ and weight matrix $\mathbf{W}_{\mathrm{sem}}^{(l)}$ compute the importance of each relation m for node i:

$$w_{i,m}^{(l)} = \mathbf{q}^{(l)\top} \tanh\left(\mathbf{W}_{\mathrm{sem}}^{(l)} \mathbf{h}_i^{(l+1,m)} + \mathbf{b}_{\mathrm{sem}}^{(l)}\right) \tag{7}$$

$$\beta_{i,m}^{(l)} = \frac{\exp\left(w_{i,m}^{(l)}\right)}{\sum_{r\in\mathcal{M}} \exp\left(w_{i,r}^{(l)}\right)} \in (0,1), \quad \sum_{m\in\mathcal{M}} \beta_{i,m}^{(l)} = 1 \tag{8}$$

The term $\beta_{i,m}^{(l)}$ represents the semantic attention weight, which quantifies the relative contribution of relation m to the embedding of OA i. The final aggregated embedding for the layer is:

$$\bar{\mathbf{h}}_i^{(l+1)} = \sum_{m\in\mathcal{M}} \beta_{i,m}^{(l)} \mathbf{h}_i^{(l+1,m)} \tag{9}$$

This fused representation $\bar{\mathbf{h}}_i^{(l+1)}$ serves as the input to the subsequent layer. The final node embedding is $\mathbf{z}_i = \bar{\mathbf{h}}_i^{(L)}$.

The heterogeneous configurations were defined as follows:

- **Model 2**: $\mathcal{M} = \{contig, 15\_min\_walk\}$
- **Model 3**: $\mathcal{M} = \{contig, 15\_min\_multi\}$
- **Model 4**: $\mathcal{M} = \{contig, 15\_min\_walk, 15\_min\_multi\}$

All encoders employed two message-passing layers ($L = 2$), aggregating information up to two hops. This constraint prevents over-smoothing, a phenomenon in which deeper GNNs produce uniform node embeddings across the graph (Rusch et al., 2023). The per-head hidden size was set to 8 and the embedding size to $d = 16$ to provide a bottleneck that is sufficient for the $F = 23$ input dimensions while keeping the latent space compact for clustering.[4] A dropout rate of 0.6 was applied to each layer as the default setting of GAT (Veličković et al., 2018) and HAN (Wang et al., 2019), to improve optimisation stability by reducing reliance on specific node or edge attributes during training.

4.5.2. Decoders and loss function

Two decoders are employed to jointly reconstruct node attributes of urban function and relation-specific structures of contiguity and accessibility between OAs, so that such information is effectively preserved in the embeddings.

Attribute reconstruction. Node attributes were reconstructed from embeddings using the Multi-Layer Perceptron (MLP) g_ψ. For attribute reconstruction, Smooth L1 loss (Girshick, 2015; Huber, 1992) was employed to address long-tailed node attributes persisting after log-scaling (see Fig. A.2 in Appendix A). Smooth L1 behaves as Mean Square Error (MSE) near zero, which stabilises optimisation, and transitions towards Mean Absolute Error (MAE) for larger errors, which reduces sensitivity to extreme values. It is defined as:

$$\mathrm{SmoothL1}(x,\hat{x}) = \begin{cases} \frac{1}{2}(x-\hat{x})^2/\delta, & \text{if } |x-\hat{x}| < \delta \\ |x-\hat{x}| - \frac{1}{2}\delta, & \text{otherwise} \end{cases} \tag{10}$$

with $\delta = 1.0$. The feature reconstruction loss is computed as:

$$\mathcal{L}_{\mathrm{feat}} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{SmoothL1}\left(\tilde{\mathbf{x}}_i, \hat{\tilde{\mathbf{x}}}_i\right). \tag{11}$$

Structure reconstruction. For each relation $m \in \mathcal{M}$, edge existence was reconstructed with a DistMult decoder (Yang et al., 2014) with learnable relation vector $\mathbf{r}_m \in \mathbb{R}^d$:

$$p_m(i,j) = \sigma\left(\mathbf{z}_i^\top \mathrm{diag}(\mathbf{r}_m)\mathbf{z}_j\right), \tag{12}$$

where $\sigma(\cdot)$ is the sigmoid function. This encourages embeddings to preserve observed connectivity under each relation while contrasting against sampled non-edges. Given observed edges $\mathcal{E}^{(m)+}$ and sampled negatives $\mathcal{E}^{(m)-}$, the structural loss is:

$$\mathcal{L}_{\mathrm{struct}}^{(m)} = -\frac{1}{|\mathcal{E}^{(m)+}|}\sum_{(u,v)\in\mathcal{E}^{(m)+}} \log p_m(u,v) - \frac{1}{|\mathcal{E}^{(m)-}|}\sum_{(u,v)\in\mathcal{E}^{(m)-}} \log\big(1-p_m(u,v)\big) \tag{13}$$

Training objective. The total loss combines feature reconstruction and relation-specific structure reconstruction:

$$\mathcal{L} = \mathcal{L}_{\mathrm{feat}} + \frac{1}{|\mathcal{M}|}\sum_{m\in\mathcal{M}} \mathcal{L}_{\mathrm{struct}}^{(m)}. \tag{14}$$

where $|\mathcal{M}|$ denotes the number of relation types.

Models were trained with a learning rate of 0.005 using the Adam optimiser (Kingma & Ba, 2014) and early stopping with a patience of 100 epochs, both following the default setting of HAN (Wang et al., 2019).

[4] With $H = 4$ multi-head attentions concatenated, the total dimension is 32.

4.6. Results

4.6.1. Clustering performance and embedding-space structure

Node embeddings of OAs were clustered using K-Means, varying the cluster count (k) from 2 to 50. For each k, the outcome with the lowest inertia across 10 random initialisations was retained. As a baseline, Principal Component Analysis (PCA) was applied to the scaled node-attribute matrix ($\tilde{\mathbf{X}}$) before K-Means, as a linear dimension reduction to the same dimensions of embeddings. In addition, structure-only variants of the GAE models (Models 1S–4S) were introduced to isolate the effect of graph topology by removing node attributes (replacing $\tilde{\mathbf{x}}_i$ with an all-ones vector $\mathbf{1}$ and setting the feature reconstruction loss $\mathcal{L}_{\mathrm{feat}} = 0$). To ensure fair comparison, learned node embeddings were L2-normalised and constrained to the unit length prior to clustering, so that Euclidean distances used by K-Means are not dominated by differences in embedding magnitude.

Clustering performance was evaluated by two complementary criteria to assess both the internal cohesion of the embeddings and their structural alignment with the underlying connections.

First, the Silhouette score (Rousseeuw, 1987) was used to assess cluster cohesion and separation in the latent space, guiding the selection of k. For a node i in cluster C_I with learned embedding $\mathbf{z}_i$, the score $s(i) \in [-1, 1]$ is defined as:

$$s(i) = \frac{b(i)-a(i)}{\max\{a(i), b(i)\}} \tag{15}$$

$$a(i) = \frac{1}{|C_I|-1}\sum_{j\in C_I, j\neq i} \|\mathbf{z}_i - \mathbf{z}_j\|$$
$$b(i) = \min_{J\neq I}\frac{1}{|C_J|}\sum_{j\in C_J} \|\mathbf{z}_i - \mathbf{z}_j\| \tag{16}$$

where $a(i)$ is the mean Euclidean distance to all other nodes in the same cluster, and $b(i)$ is the minimum mean distance to nodes in any other cluster. The overall score is the mean $s(i)$ across all nodes $i \in \mathcal{V}$. Since the Euclidean distance between normalised embeddings corresponds to angular similarity with the same scale between −1 and 1 across all model settings, the Silhouette score provides a robust geometric measure for evaluating how well the embeddings could be grouped as clusters.

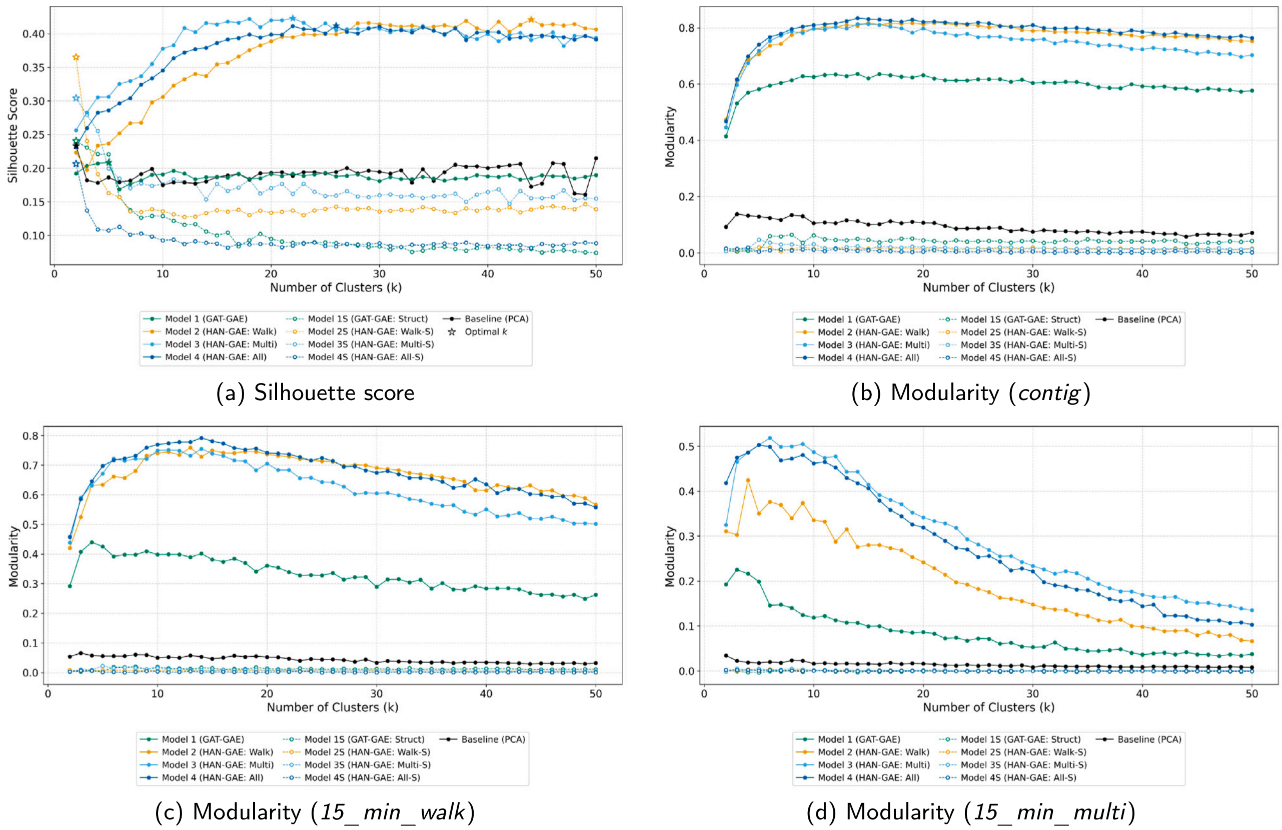
As another internal validation criterion, modularity (Newman, 2006) was introduced to measure how well the clusters reflect the connections between OAs. For each k, the K-Means partition was treated as a candidate community assignment, and its modularity was computed separately for each relation $m \in \mathcal{M}$. Given the adjacency matrix $\mathbf{A}^{(m)}$, the degree $k_i^{(m)} = \sum_j \mathbf{A}_{ij}^{(m)}$, and the total number of edges $|\mathcal{E}^{(m)}| = \frac{1}{2}\sum_{i,j}\mathbf{A}_{ij}^{(m)}$, the relation-specific modularity $Q^{(m)} \in [-1, 1]$ is calculated as:

$$Q^{(m)} = \frac{1}{2|\mathcal{E}^{(m)}|}\sum_{i,j}\left(A_{ij}^{(m)} - \frac{k_i^{(m)}k_j^{(m)}}{2|\mathcal{E}^{(m)}|}\right)\delta(c_i, c_j) \tag{17}$$

where c_i denotes the cluster assignment of node i, and $\delta(c_i, c_j)$ equals 1 if $c_i = c_j$, and 0 otherwise. This relation-wise modularity measures how well clusters align with the connectivity patterns of each underlying graph.

The Silhouette scores in the left part of Table 3 and Fig. 14(a) reveal a clear separation between model families. Models 2, 3, and 4 achieved high scores, demonstrating that incorporating accessibility-based relations markedly improves embedding separability for the clustering of OAs. By contrast, the baseline and Model 1 scored low, indicating limited capacity to resolve fine-grained functional differences. The structure-only variants (Models 1S–4S) failed to produce spatially coherent clusters, as confirmed in Fig. 15(a), underscoring that node attributes encoding urban function are essential: graph structure alone does not recover meaningful functional partitions.

The modularity analysis in the right part of Table 3 and Figs. 14(b)–(d) clarifies which relations align most consistently with the



(a) Silhouette score (b) Modularity (*contig*)

(c) Modularity (*15_min_walk*) (d) Modularity (*15_min_multi*)

Fig. 14. Silhouette scores and Modularity for varying cluster counts (k = 2–50). The baseline and Models 1–4 are drawn as solid lines and the structure-only variants (Models 1S–4S) as dashed lines with white-filled markers in the same colour.

Table 3
Clustering performance metrics (Silhouette score and Modularity) across varying cluster counts (k). Values denote mean ± standard deviation recorded across the range $k \in [2, 50]$.

Model	Silhouette score Mean ± Std	Modularity (*contig*) Mean ± Std	Modularity (*15_min_walk*) Mean ± Std	Modularity (*15_min_multi*) Mean ± Std
Baseline (PCA)	0.19 ± 0.01	0.09 ± 0.02	0.04 ± 0.01	0.01 ± 0.01
Model 1 (GAT-GAE)	0.19 ± 0.01	0.60 ± 0.03	0.33 ± 0.05	0.08 ± 0.05
Model 2 (HAN-GAE: Walk)	0.37 ± 0.06	0.77 ± 0.06	0.67 ± 0.07	0.20 ± 0.10
Model 3 (HAN-GAE: Multi)	0.39 ± 0.04	0.75 ± 0.06	0.62 ± 0.09	0.30 ± 0.13
Model 4 (HAN-GAE: All)	0.38 ± 0.04	0.79 ± 0.06	0.68 ± 0.07	0.28 ± 0.14
Model 1S (GAT-GAE: Struct)	0.10 ± 0.04	0.04 ± 0.01	0.01 ± 0.00	0.00 ± 0.00
Model 2S (HAN-GAE: Walk-S)	0.15 ± 0.04	0.01 ± 0.00	0.01 ± 0.00	0.00 ± 0.00
Model 3S (HAN-GAE: Multi-S)	0.17 ± 0.03	0.02 ± 0.01	0.01 ± 0.00	0.00 ± 0.00
Model 4S (HAN-GAE: All-S)	0.09 ± 0.02	0.01 ± 0.00	0.00 ± 0.00	0.00 ± 0.00

Note: Models 1S–4S are structure-only variants in which the node attribute matrix $\tilde{\mathbf{X}}$ is replaced by an all-ones matrix ($\mathbf{1} \in \mathbb{R}^{N\times 23}$), isolating the contribution of graph structure from node attributes.

inferred clusters. Regarding spatial contiguity in (b), heterogeneous models achieved higher modularity than Model 1, which still exceeded the baseline. Regarding walk-based and multimodal accessibility in (c) and (d), models that introduced the relevant relations showed higher modularity (e.g., Models 2 and 4 for walk-based accessibility, and Models 3 and 4 for multimodal accessibility). This indicates that embedding similarity preserved the connectivity patterns implied by each model's assumed relations. The consistently low modularity of the structure-only models (1S–4S) across all relations confirms that functional communities arise from the combination of node attributes and accessibility relations, not from topology alone.

Together, the Silhouette and modularity results indicate that the models yield clusters that are both well separated in the latent space and structurally coherent under the underlying OA relations. The cluster maps at the optimal k (Fig. 15(a)) illustrate this clearly. Compared to the baseline, Model 1 recovered spatial clusters including the city centre in the north-west. Model 2 captured small-scale clusters at the walking-catchment scale, whereas Model 3 delineated broader clusters corresponding to bus-accessible areas. Model 4 produced intermediate clusters encompassing both walking-only and bus-accessible zones. The structure-only models, by contrast, failed to produce spatially coherent clusters. A supporting robustness check is provided in Appendix C.

To evaluate how each model encodes spatial and functional proximity to the urban core, cosine similarity was computed between the embedding of the OA containing Liverpool ONE and those of all other OAs. Liverpool ONE, a major retail and commercial centre with an integrated bus terminal, provides an intuitive reference point for interpreting how models capture proximity to the economic core under different transport assumptions. For two node embeddings $\mathbf{z}_i$ and $\mathbf{z}_j$,

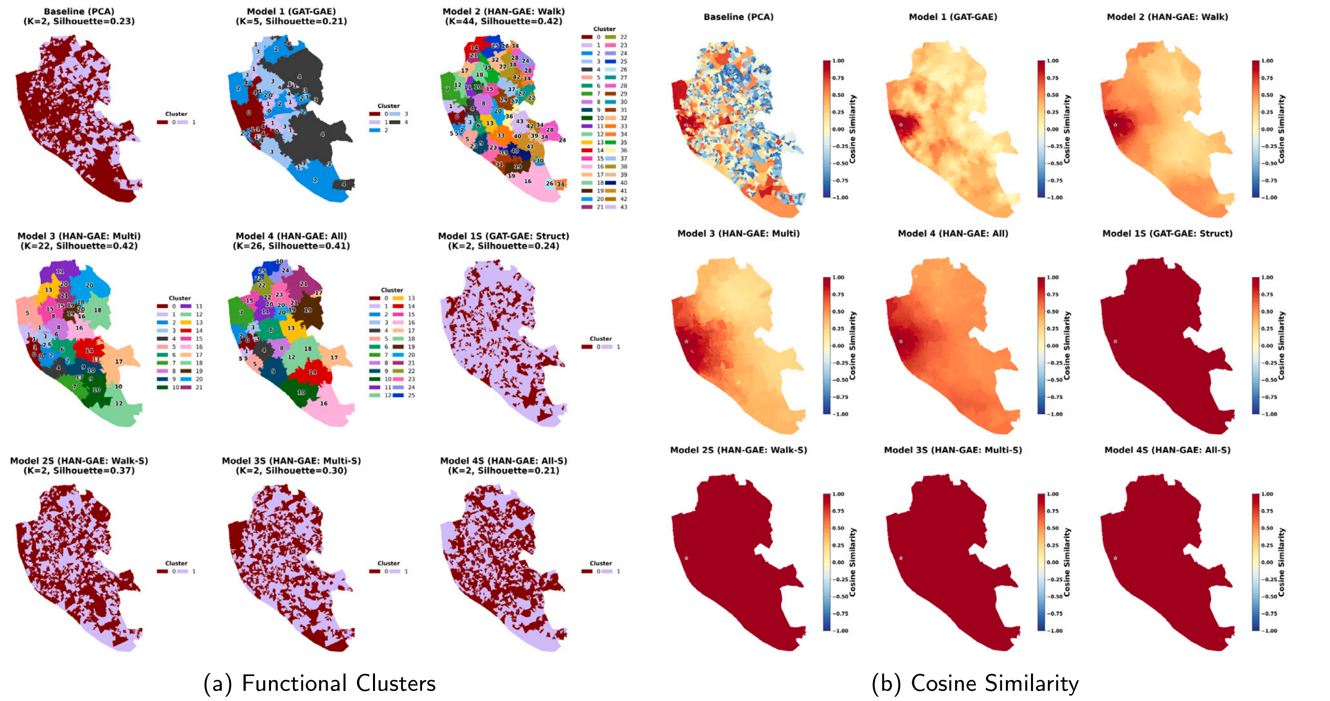


Fig. 15. Spatial distribution of (a) functional clusters and (b) cosine similarity of embeddings relative to Liverpool ONE (53.403553, −2.986844) across the baseline, proposed (1–4), and structure-only (1S–4S) models. In (a), clusters are numbered in descending order of the cosine similarity between their centroid in the embedding space and that of the cluster containing the OA of Liverpool ONE (ID: E00176629), so that cluster 0 is the cluster containing that OA. The numbers are given in the legend of each panel and, for Models 1–4, are also placed on the mapped clusters. In (b), each OA is coloured by the cosine similarity between the embedding of the OA containing Liverpool ONE (ID: E00176629) and the embedding of that OA.

cosine similarity is defined as $\cos(\mathbf{z}_i, \mathbf{z}_j) = \mathbf{z}_i^\top \mathbf{z}_j \,/\, (\|\mathbf{z}_i\|\,\|\mathbf{z}_j\|) \in [-1, 1]$, measuring directional alignment in the embedding space. This metric reflects how similar an OA's context is to that of the city centre in terms of both urban-function composition and the connectivities defined. As embeddings are normalised in advance ($\|\mathbf{z}_i\| = 1$), the measured cosine similarity is equivalent to their inner product $\mathbf{z}_i^\top \mathbf{z}_j$.

Fig. 15(b) maps the cosine similarity between the Liverpool ONE embedding and all other OA embeddings for each model. The baseline captured similarity driven by the city centre's rich and diverse urban functions, with OAs lacking such functions scoring markedly lower. Models 1–4, by contrast, exhibited spatial smoothing. OAs with dissimilar urban functions gained higher similarity through relational context with their neighbours. Each heterogeneous model displayed a distinct smoothing pattern reflecting its accessibility assumptions. Model 2 maintained high similarity within the 15-minute walking catchment, with rapid decay beyond that range. Model 3 extended similarity with the longer distance connectivity by bus routes. Model 4 concentrated high similarity within the walking catchment while retaining moderate similarity with suburban areas served by transit. This smoothing effect is consistent with previous findings on the role of public transport in reducing inequality in access to urban functions (Guo & Brakewood, 2024; Hernandez, 2018).

4.6.2. Analysis of cosine similarity decay

To understand how the obtained embeddings align with physical and multimodal accessibility, cosine similarity decay relative to the city centre (Liverpool ONE) was measured by travel time distances. Using the `create_isochrone()` function, the population-weighted centroids of OAs were grouped by walking and multimodal travel times (Fig. 16), providing a structured framework to interpret functional similarity in terms of transit and pedestrian accessibility. Based on the resulting isochrones, mean cosine similarity to the central OA was computed for each bin, as presented in Fig. 17. For the binning interval, 5-minute travel time was employed up to 30 min, which is the limit of message-passing by 2-layer GAT and HAN. By comparing the decay processes across models, we examined how similarity to the city centre propagated through the embeddings depending on the accessibility assumed by each model's architecture.

Fig. 17 shows how the cosine similarity of node embeddings decays across travel time bins. A clear separation emerges between the proposed models (Models 1–4) and the structure-only variants (Models 1S–4S). The latter maintain an almost constant similarity of approximately 1.0 across all bins, implying that topology alone does not capture urban spatial heterogeneity and that node attributes are required to recover the expected distance-decay effect. The proposed four models also sustain cosine similarity compared to the baseline with PCA, which sharply loses spatial coherence and yields negative similarity for multimodal trips exceeding 15 min. In the walk-based isochrones, Models 1 to 4 keep high similarity up to 15 min. The multimodal isochrones, by contrast, highlight the advantage of the heterogeneous architecture. Model 1 decays quickly for longer multimodal trips, whereas Models 3 and 4 retain higher similarity, suggesting that incorporating transit layers helps preserve relationships between geographically distant locations that are functionally connected via transit.

4.6.3. Model interpretation

To investigate how the HAN-GAE models prioritise the different spatial relations during the learning process, the semantic attention weights ($\beta_{i,m}^{(l)}$) were visualised for each OA in Fig. 18, separately for the two message-passing layers ($l = 1$ and $l = 2$). The two layers exhibit distinct roles in relational encoding. In the first layer shown in Fig. 18(a), models prioritise accessibility relations (*15_min_walk* and *15_min_multi*) while assigning lower weights to spatial contiguity (*contig*). Conversely, in the second layer shown in Fig. 18(b), *contig* obtains attention weights across all models. This suggests a two-step aggregation process, where models initially draw on broad network reachability to capture functional similarity, and subsequently employ the contiguity to smooth them into spatially cohesive embeddings as seen in Fig. 15.

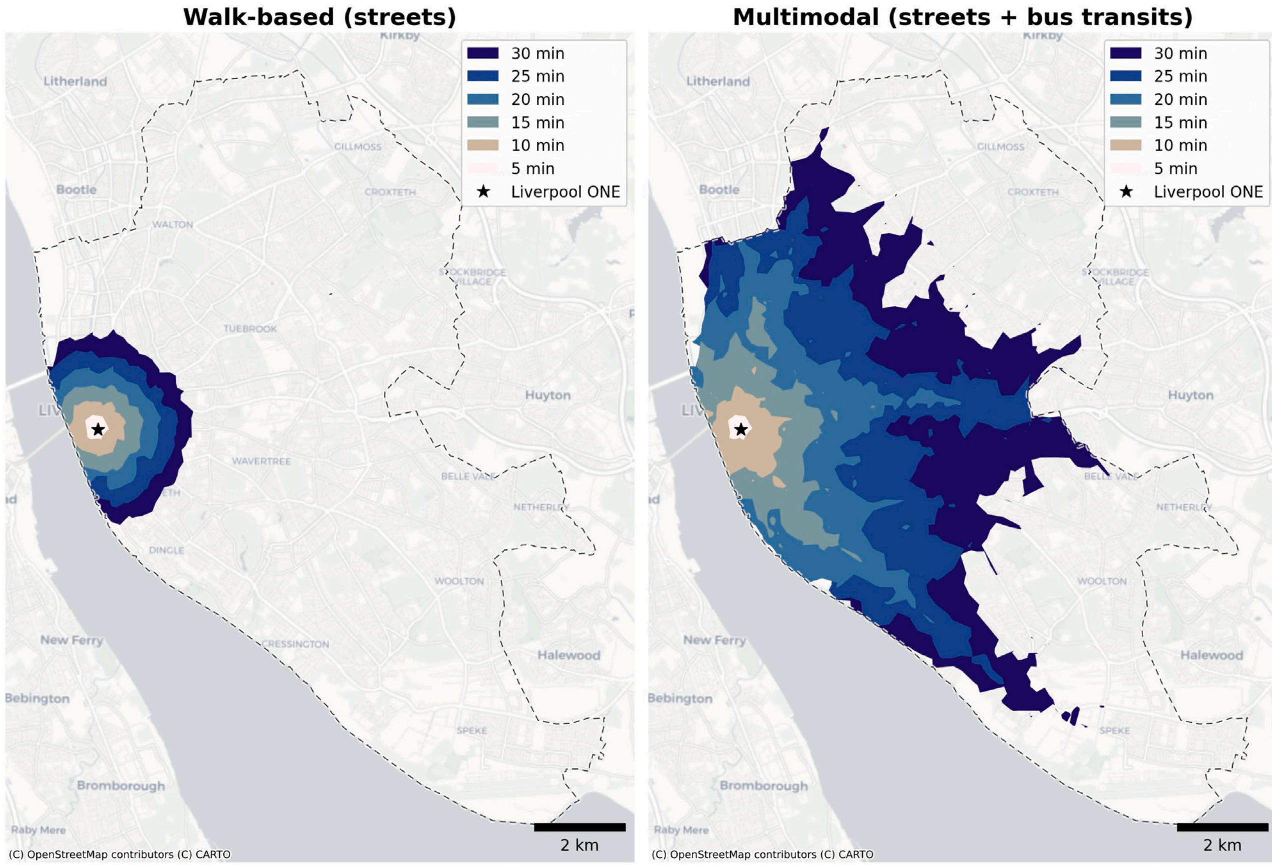


Fig. 16. Isochrones from Liverpool ONE (53.403553, −2.986844) generated by `create_isochrone()`. The left subplot shows walk-only isochrones based solely on street segments, while the right subplot displays multimodal isochrones integrating street segments and bus transit.

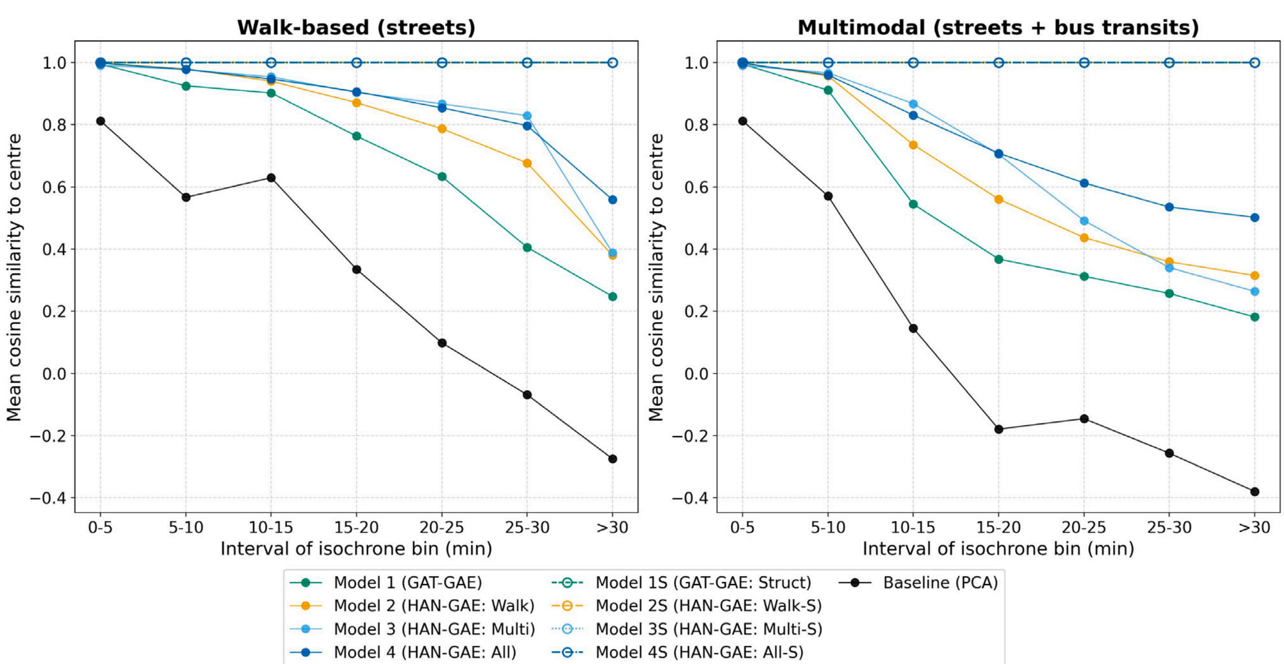
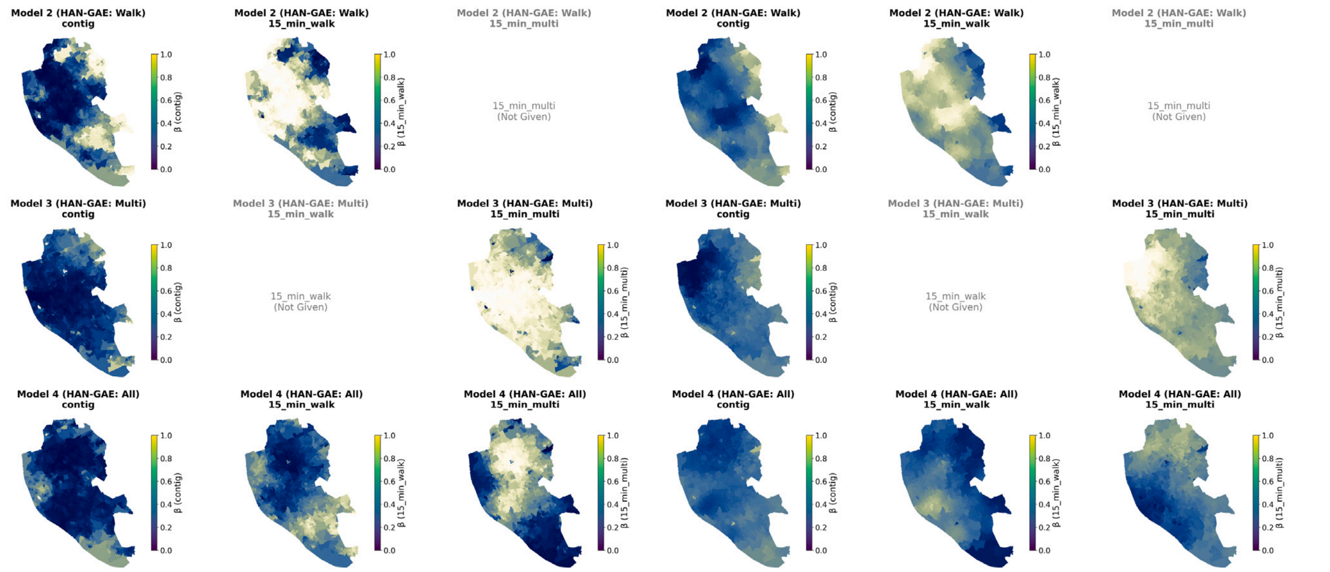


Fig. 17. Cosine similarity decay of embeddings relative to the OA containing Liverpool ONE (ID: E00176629). Mean cosine similarity of embeddings to the central OA is computed for each travel time bin.



(a) Layer 1 ($l = 1$) (b) Layer 2 ($l = 2$)

Fig. 18. Spatial distribution of learned semantic attention weights ($\beta_{i,m}^{(l)}$) for (a) Layer 1 ($l = 1$) and (b) Layer 2 ($l = 2$). Rows correspond to Models 2–4, and columns to relations: *contig*, *15_min_walk*, and *15_min_multi*.

Model 4, which integrates all three relations, distinguishes the roles of the two accessibility modes. In both layers, the attention weights for *15_min_walk* are higher in the north-western city centre than those for *15_min_multi*. This contrast indicates that the model values the information of neighbouring OAs differently by location, where the city centre prioritises the walkable area nearby, and the suburb is characterised by the middle-distance neighbours from transit networks.

5. Discussion

City2Graph addresses a methodological gap in urban graph representation learning by providing a standardised pipeline that converts spatial data into graph structures and GNN-ready tensors while preserving coordinate reference systems, stable identifiers, and geometries. By reducing reliance on bespoke conversions between GeoPandas, NetworkX, and PyTorch Geometric, the library lowers the risk of misaligned spatial metadata and improves reproducibility. Importantly, its bidirectional conversions support post hoc interpretation by allowing embeddings and predictions to be mapped back to their original spatial units for visualisation and validation within established spatial analytical workflows.

Another contribution is a consistent data model for constructing heterogeneous urban graphs and composing relations across domains. This enables joint modelling of built form and network configuration, multimodal accessibility derived from pedestrian networks and GTFS schedules, and mobility relations derived from OD flows. The proximity tools further support inter-layer coupling, including bridging nodes across different layers and grouping point nodes within polygon nodes of another layer, which is useful for catchment, accessibility, and service provision analyses.

Rather than operating in isolation, City2Graph fosters the open urban data science ecosystem by acting as connective infrastructure between established data sources and tools. The library amplifies the research value of open data sources and standard formats (e.g., OSM, Overture Maps, GTFS) by streamlining their integration into GNN workflows whilst leveraging existing libraries such as OSMnx and libpysal. To ensure long-term sustainability, the library adopts a permissive BSD 3-Clause License and open GitHub governance to encourage broad adoption and community-driven maintenance.

However, the library may face scalability constraints arising from two aspects, as of version 0.4.0. Algorithmically, certain operations, particularly shortest-path computation for metapath projection and pairwise distance calculation for proximity graphs, carry inherent costs that grow substantially with graph size. These costs would persist regardless of implementation. At the implementation level, the library currently processes graphs in memory and relies on Python-level iteration in several modules (e.g., network distance computation in the proximity module). While the graph conversion already employs vectorised operations and sparse matrix methods, extending the overall pipeline to regional or national scales would further benefit from batch processing, memory-efficient data structures, or integration with distributed computation frameworks and graph databases.

Temporal representation is a further limitation. Although City2Graph can ingest time-dependent sources such as transit schedules, the resulting graphs are typically constructed as static snapshots or aggregated summaries, which constrains the representation of peak variability, disruption, and seasonal change. Addressing this will require explicit temporal abstractions, such as time-stamped edges and evolving attributes with PyTorch Geometric Temporal (Rozemberczki et al., 2021), coupled with robust data versioning to support comparable longitudinal analyses. Priorities for future development are therefore to accelerate core spatial graph operations while enabling chunked, identifier-preserving construction, and to extend the library with native support for dynamic heterogeneous graphs.

6. Conclusion

While heterogeneous GNNs represent a promising modelling framework for identifying multi-layered urban complexity, their application has been constrained by fragmented tool ecosystems. City2Graph provides a unified pipeline for constructing heterogeneous graphs from multiple domains of morphology, transportation, mobility, and proximity. Being compatible between GeoPandas, NetworkX, and PyTorch Geometric, City2Graph allows users to switch seamlessly across spatial analysis, network analysis, and GNN workflows while preserving coordinate reference systems and stable identifiers, so that embeddings can be mapped back to their original spatial units for interpretation. In addition, the library supports the construction of metapaths as higher-order spatial relations, defining semantic connections across multiple

Table A.1
Summary statistics of urban function as node attributes (raw and scaled).

Name	Raw				Scaled			
	Min	Mean	Max	Std	Min	Mean	Max	Std
arts_culture	0.0	0.27	33.00	1.41	−0.32	0.0	9.26	1.0
automotive_facility	0.0	0.62	53.00	2.41	−0.46	0.0	7.09	1.0
consumer_service	0.0	1.58	86.00	4.23	−0.71	0.0	5.21	1.0
corporate_service	0.0	1.70	120.00	7.26	−0.62	0.0	5.82	1.0
education	0.0	0.66	32.00	1.84	−0.58	0.0	6.11	1.0
entertainment	0.0	0.51	47.00	2.24	−0.42	0.0	7.69	1.0
food_beverage	0.0	2.23	240.00	9.43	−0.66	0.0	5.81	1.0
healthcare	0.0	0.92	55.00	2.84	−0.58	0.0	5.90	1.0
hotel_lodging	0.0	0.32	27.00	1.46	−0.34	0.0	8.00	1.0
industrial_service	0.0	0.87	58.00	2.67	−0.64	0.0	6.46	1.0
park	0.0	0.13	6.00	0.46	−0.30	0.0	7.40	1.0
public_service	0.0	0.51	15.00	1.32	−0.51	0.0	5.20	1.0
religion	0.0	0.27	7.00	0.73	−0.42	0.0	5.32	1.0
retail	0.0	2.25	271.00	9.91	−0.71	0.0	6.04	1.0
sports_fitness	0.0	0.41	25.00	1.33	−0.44	0.0	6.88	1.0
transportation_facility	0.0	0.38	34.00	1.71	−0.40	0.0	8.13	1.0
land_use_green_space	0.0	12 596.43	1 486 093.29	58 398.57	−1.01	0.0	2.25	1.0
land_use_residential	0.0	33 616.78	205 941.04	20 749.60	−5.61	0.0	1.25	1.0
land_use_industrial	0.0	5814.28	1 540 174.93	57 100.26	−0.38	0.0	4.43	1.0
land_use_public_services	0.0	4300.38	432 561.96	21 592.67	−0.53	0.0	2.98	1.0
land_use_transportation	0.0	789.81	69 901.94	4631.77	−0.41	0.0	3.72	1.0
land_use_commercial	0.0	2864.61	281 235.37	13 934.25	−0.53	0.0	3.07	1.0
land_use_agricultural	0.0	950.24	1 030 935.31	26 522.51	−0.11	0.0	14.91	1.0

Note: All land use attributes (`land_use_green_space`, `land_use_residential`, `land_use_industrial`, `land_use_public_services`, `land_use_transportation`, `land_use_commercial`, `land_use_agricultural`) represent area in m^2 aggregated to each OA, while the remaining attributes are expressed as counts of POIs.

node and edge types, such as multimodal accessibility. An empirical demonstration of functional clustering in Liverpool confirmed the efficacy of this approach, as heterogeneous models utilising accessibility-based metapaths outperformed homogeneous models limited to spatial contiguity. Future development will prioritise improved computational scalability and support for dynamic temporal graphs.

CRediT authorship contribution statement

Yuta Sato: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Elisabetta Pietrostefani:** Writing – review & editing, Supervision. **Ron Mahabir:** Writing – review & editing, Supervision. **Daniel Arribas-Bel:** Writing – review & editing, Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Gemini 3.1 Pro in order to correct grammar and proofread the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplemental information on node and edge attributes in the case study

Table A.1 presents the summary statistics of the OA-level node attributes of urban functions used for the case study, reported both in raw form and after the log-transform and standardisation described in Section 4.4. Figs. A.1 and A.2 show the corresponding distributions as histograms. Table A.2 presents the summary statistics of edge counts and edge weights.

Appendix B. Model details

This appendix provides the full mathematical definitions of the attention-based encoders and semantic attention used in Section 4.5.

B.1. Edge-aware node attention (GAT)

For relation $m \in \mathcal{M}$, let $\mathcal{N}_m(i) = \{j : (i,j) \in \mathcal{E}^{(m)}\}$ denote the neighbourhood of node i. Let $\mathbf{h}_i^{(l)} \in \mathbb{R}^{d_l}$ denote the node representation at layer $l \in \{0, \dots, L\}$, which serves as the shared input for all relations. For the input layer, $\mathbf{h}_i^{(0)} = \tilde{\mathbf{x}}_i$, with $d_0 = F$ and $d_L = d$. H attention heads are used and aggregated by concatenation. For each head $h \in \{1, \dots, H\}$, the attention score and coefficient are given by

$$e_{ij}^{(l,m,h)} = \mathrm{LeakyReLU}\left(\mathbf{a}^{(l,m,h)\top}\left[\mathbf{W}^{(l,m,h)}\mathbf{h}_i^{(l)} \,\|\, \mathbf{W}^{(l,m,h)}\mathbf{h}_j^{(l)} \,\|\, \mathbf{U}^{(l,m,h)}\tilde{\tau}_{ij}^{(m)}\right]\right) \tag{18}$$

$$\alpha_{ij}^{(l,m,h)} = \frac{\exp\left(e_{ij}^{(l,m,h)}\right)}{\sum_{k\in\mathcal{N}_m(i)\cup\{i\}}\exp\left(e_{ik}^{(l,m,h)}\right)}. \tag{19}$$

where $\|$ denotes concatenation, and $\mathbf{W}^{(l,m,h)}$, $\mathbf{U}^{(l,m,h)}$, and $\mathbf{a}^{(l,m,h)}$ are learnable parameters (with shapes determined by the layer dimensions). The layer update aggregates neighbour messages within each

Table A.2
Summary statistics of edge counts and edge weights in the case study.

Edge type	Edge count	Edge weight			
		Min	Mean	Max	Std
('oa', 'is_nearby', 'street_connector')	1624	0.61	23.81	113.46	16.74
('street_connector', 'is_connected_to', 'street_connector')	65 386	0.01	32.76	1081.63	38.96
('bus_station', 'is_nearby', 'street_connector')	1724	0.65	15.76	99.85	10.10
('bus_station', 'is_next_to', 'bus_station')	2290	10.00	92.65	1980.00	93.48
('oa', 'contig', 'oa')	4782	23.33	225.89	1699.17	151.71
('oa', '15_min_walk', 'oa')	28 265	30.55	620.23	899.98	195.94
('oa', '15_min_multi', 'oa')	69 403	232.28	766.26	900.00	100.14

Note: Edge weights for `('oa', 'is_nearby', 'street_connector')`, `('bus_station', 'is_nearby', 'street_connector')`, and `('oa', 'contig', 'oa')` are calculated from Euclidean distance between nodes divided by 4.8 km/h and the others are measured as travel times in seconds for network-derived relations. Edge counts are reported per undirected pair (one row per pair, as returned by `pyg_to_gdf()`), whereas the underlying `edge_index` stores both directions and thus contains twice the pair count.

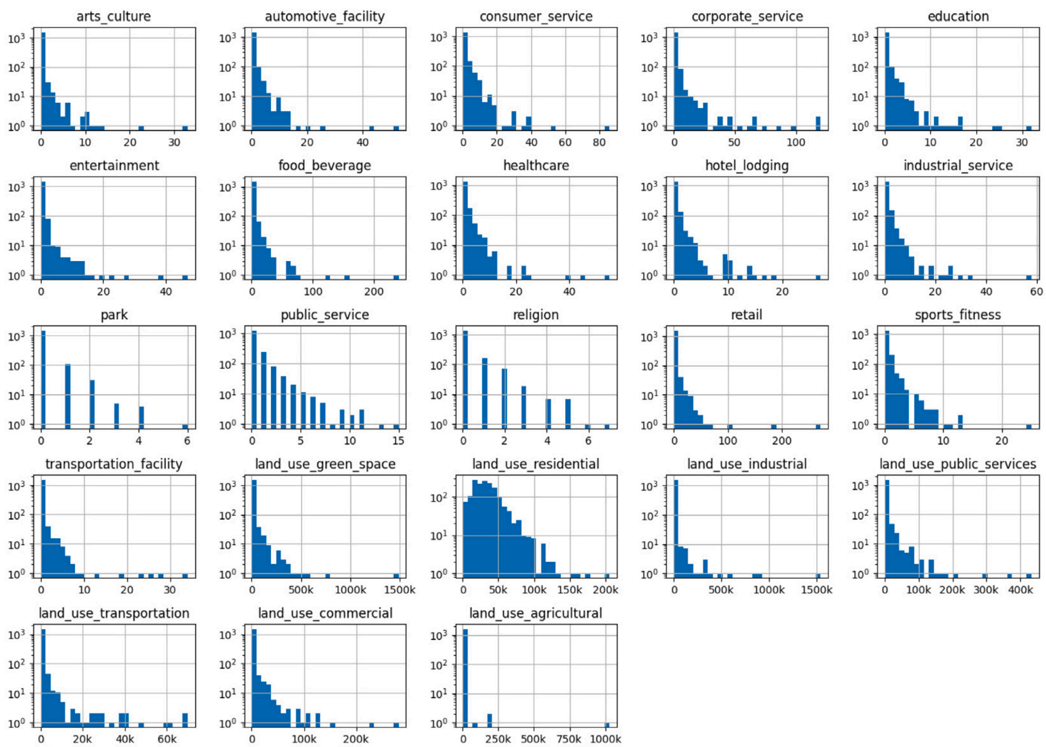


Fig. A.1. Distributions of raw node attributes in the case study.

head and concatenates across heads

$$\mathbf{h}_i^{(l+1,m)} = \Big\Vert_{h=1}^{H} \sum_{j\in\mathcal{N}_m(i)\cup\{i\}} \alpha_{ij}^{(l,m,h)}\, \mathbf{W}^{(l,m,h)} \mathbf{h}_j^{(l)} \in \mathbb{R}^{d_{l+1}}. \tag{20}$$

An ELU activation is applied to each GAT output. In the heterogeneous encoder (Models 2–4), semantic attention (described in Section 4.5) fuses the relation-specific outputs at each layer.

Appendix C. Supplemental clustering evaluation

To verify that the clustering results are not due to a specific algorithm or metric, we applied three complementary validation approaches: Calinski–Harabasz and Davies–Bouldin indices (Caliński & Harabasz, 1974; Davies & Bouldin, 1979) for K-Means across $k \in [2, 50]$, HDBSCAN (Campello et al., 2013) evaluated with Density-Based Clustering Validation (DBCV), and t-SNE manifold visualisation. For HDBSCAN, minimum cluster size was varied between 5 (default value) and 30 via grid search, ranging from localised niches to district-level communities, where each cluster has at least 1.8% of all OAs.

The results shown in Table C.1 and Figs. C.1–C.3 corroborate the main analysis. On average, Models 2–4 achieved higher Calinski–Harabasz indices, lower Davies–Bouldin indices, and higher DBCV scores than the baseline and Model 1, confirming better-defined clusters across both partitional and density-based approaches. Model 1 reached an optimal DBCV of 0.27 with 19 clusters (Fig. C.2), while its mean DBCV across the grid search remained low (0.07). Models 2–4 identified 14 to 32 finer-grained, spatially coherent functional zones while sustaining optimal DBCVs of 0.27–0.38. The t-SNE projections in Fig. C.3 reinforce this pattern, with well-separated clusters for Models 2–4 contrasting the poorly resolved embeddings of the baseline and Model 1. All structure-only models (1S–4S) exhibited low optimal DBCV scores (0.00–0.10) with severe over-segmentation or excessive noise, confirming that node attributes are essential for the proposed GAE models to capture urban functional zones.

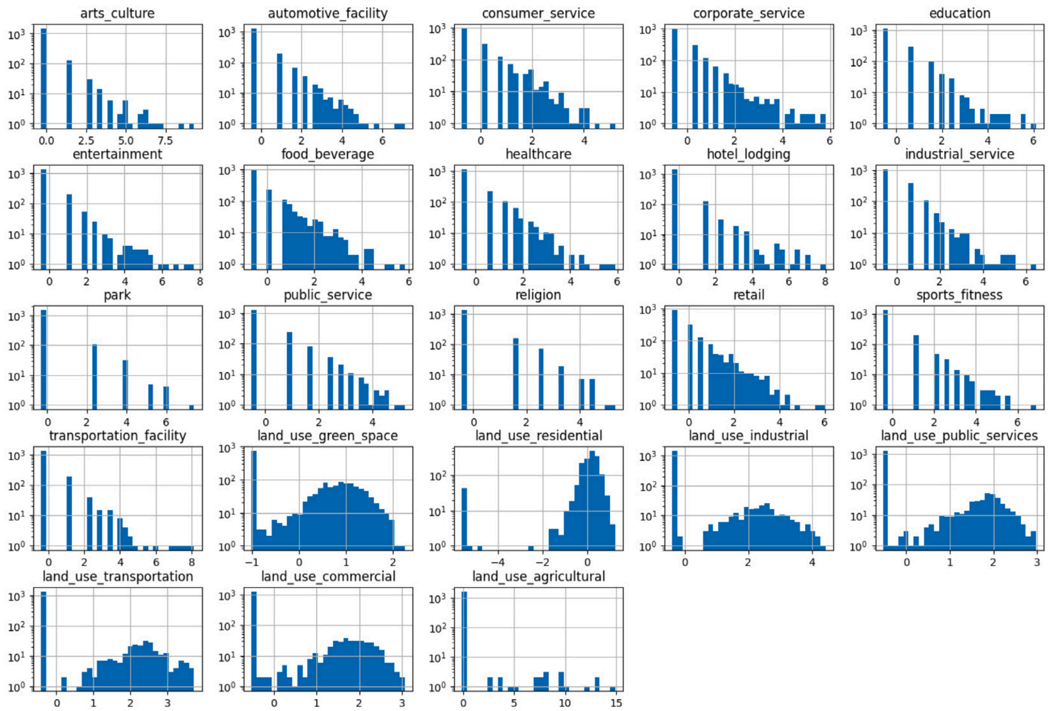
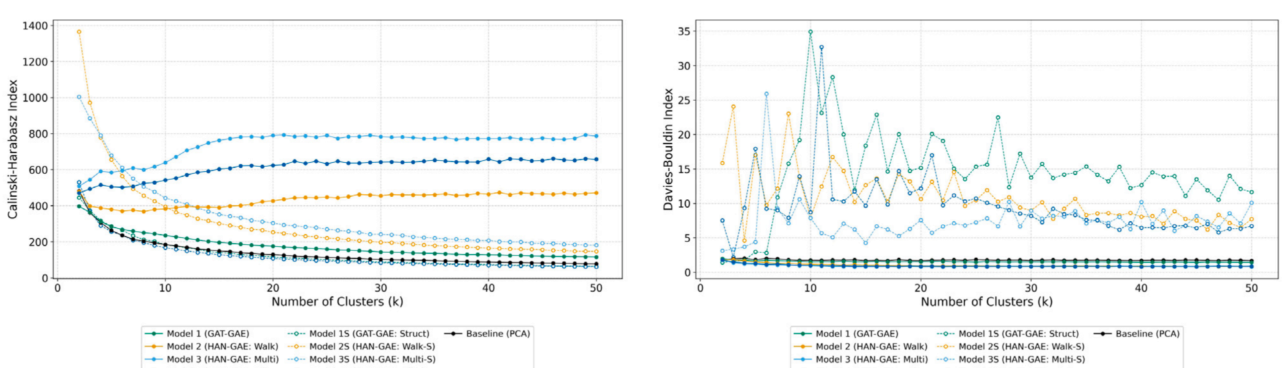


Fig. A.2. Distributions of scaled node attributes in the case study.

Table C.1
Supplemental clustering performance metrics for K-Means and HDBSCAN.

Model	K-Means (k = 2–50)		HDBSCAN			
	Calinski–Harabasz Mean ± Std	Davies–Bouldin Mean ± Std	DBCV Mean ± Std	Modularity (*contig*) Mean ± Std	Modularity (*15_min_walk*) Mean ± Std	Modularity (*15_min_multi*) Mean ± Std
Baseline (PCA)	139.06 ± 78.94	1.76 ± 0.08	0.10 ± 0.04	0.26 ± 0.06	0.17 ± 0.05	0.06 ± 0.02
Model 1 (GAT-GAE)	177.61 ± 64.97	1.54 ± 0.10	0.07 ± 0.08	0.53 ± 0.13	0.53 ± 0.10	0.28 ± 0.07
Model 2 (HAN-GAE: Walk)	434.85 ± 35.09	1.01 ± 0.25	0.20 ± 0.06	0.91 ± 0.01	0.84 ± 0.04	0.24 ± 0.08
Model 3 (HAN-GAE: Multi)	737.96 ± 76.63	0.90 ± 0.17	0.20 ± 0.08	0.86 ± 0.02	0.77 ± 0.07	0.45 ± 0.14
Model 4 (HAN-GAE: All)	611.88 ± 54.47	0.93 ± 0.19	0.19 ± 0.08	0.84 ± 0.23	0.78 ± 0.23	0.37 ± 0.13
Model 1S (GAT-GAE: Struct)	130.35 ± 84.44	14.55 ± 6.19	0.02 ± 0.00	0.01 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Model 2S (HAN-GAE: Walk-S)	298.83 ± 230.01	10.81 ± 3.87	0.03 ± 0.04	0.02 ± 0.02	0.00 ± 0.01	0.00 ± 0.00
Model 3S (HAN-GAE: Multi-S)	332.12 ± 186.49	7.46 ± 3.18	0.01 ± 0.01	0.01 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Model 4S (HAN-GAE: All-S)	123.63 ± 87.97	9.52 ± 4.46	0.00 ± 0.00	0.00 ± 0.01	0.00 ± 0.00	0.00 ± 0.00

Note: Results reflect a grid search of k between 2 and 50 for K-Means, and minimum cluster size between 5 and 30 for HDBSCAN. For HDBSCAN's modularity, noise points (cluster −1) were excluded from the calculation, and the metric was computed on the relation-specific subgraphs consisting solely of valid nodes.



(a) Calinski-Harabasz Index (b) Davies-Bouldin Index

Fig. C.1. (a) Calinski–Harabasz Index and (b) Davies–Bouldin Index for varying cluster counts (k = 2–50). As in Fig. 14, the baseline and Models 1–4 are drawn as solid lines and the structure-only variants (Models 1S–4S) as dashed lines with white-filled markers in the same colour.

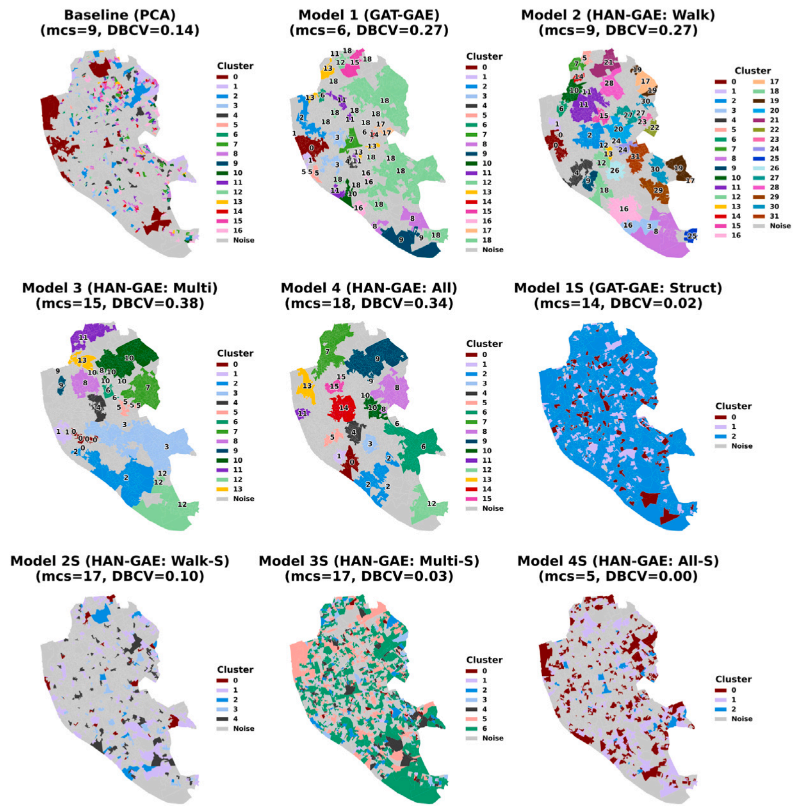
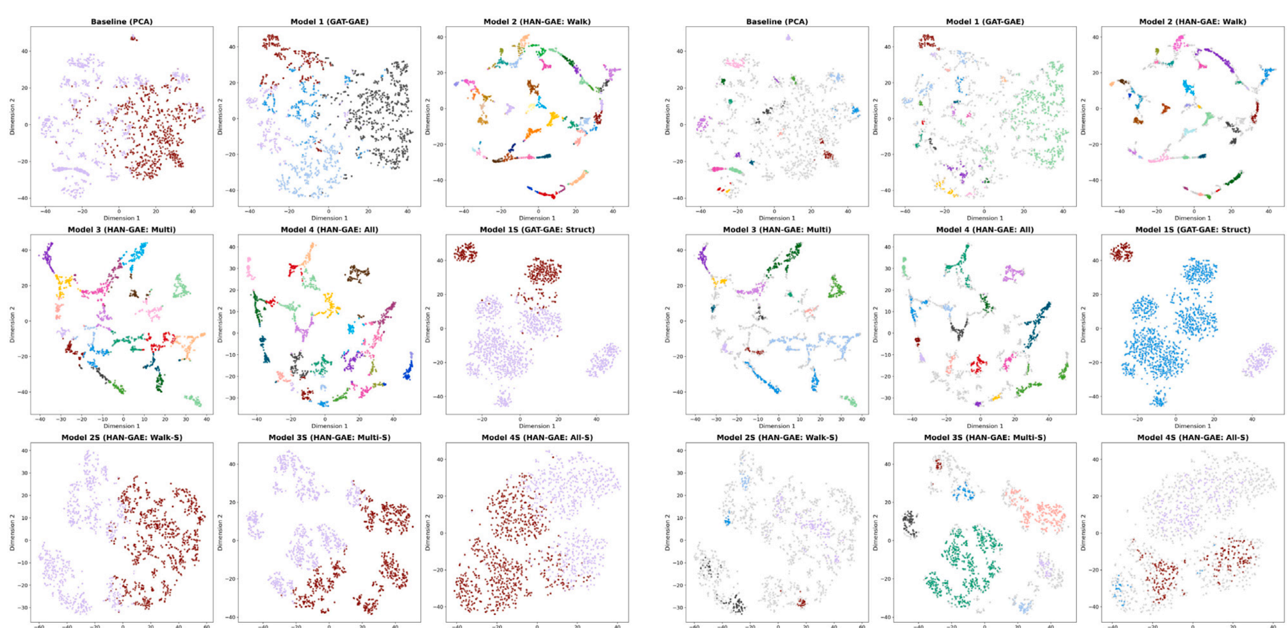


Fig. C.2. Spatial distribution of functional clusters using HDBSCAN. Noise points are shown in grey. mcs and DBCV denote minimum cluster size and Density-Based Clustering Validation, respectively.



(a) K-Means (b) HDBSCAN

Fig. C.3. Two-dimensional t-SNE projections of the node embeddings generated by the baseline and proposed models. Each point represents an Output Area (OA), coloured by the cluster assignments at (a) the optimal Silhouette score of K-Means and (b) the optimal DBCV of HDBSCAN.

Data availability

The source code of City2Graph is available at https://github.com/c2g-dev/city2graph. The source code, notebooks, and configuration files used for the case study are available at https://github.com/c2g-dev/city2graph-case-study. The dataset of the case study (inputs and derived artefacts, including features, graphs, embeddings, model parameters, clusters, figures, and tables) is archived on Zenodo at https://doi.org/10.5281/zenodo.18396285(CC BY 4.0).

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